

Senior High School



Lesson Exemplar in General Mathematics

Quarter 2

Unit

4

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Lesson Exemplar for General Mathematics
Quarter 2: Unit 4

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Republic of the Philippines
Department of Education
BUREAU OF LEARNING DELIVERY

LESSON EXEMPLAR			
Learning Area:	GENERAL MATHEMATICS		Grade Level: 11
Semester:	FIRST	Quarter: SECOND	Unit: Four

I. OBJECTIVES

Content Standard	The learners demonstrate knowledge and understanding of measurement and conversion. <i>Contextualized</i> The learners demonstrate knowledge and understanding of measurement and conversion in everyday tasks and activities, helping them be accurate, creative, and practical.
Performance Standard	By the end of the quarter, the learners are able to present solutions concerning real-life problems that require measurements and conversion (such as designing a garden, cooking recipes, or building a model) and function models. <i>Contextualized</i> By the end of the quarter, the learners are able to present solutions concerning real-life problems that require measurements and conversion (such as designing a garden, cooking recipes, or building a model) and function models by creating garden designs, blueprints, maps, sketches, recipes, travel plans, and daily routine schedules for personal and business transactions.
Learning Competencies	The learners... - determine and apply the appropriate metric units for various contexts such as units for small and large quantities and scientific measurements. - convert measurements with different units in the same system and between different systems. - determine the surface area of three-dimensional objects such as prisms, pyramids, cylinders, cones, spheres and combinations of these. - calculate the volume of prisms, pyramids, cylinders, cones, spheres and combinations of these. - Use a scale to interpret and compare distances on maps, blueprints, and sketches.

	6. solve problems involving conversion of currencies. 7. apply time conversions in various real-world scenarios such as travel planning, project management, and daily routine schedules for personal and business transactions. 8. apply temperature conversions to real-world scenarios such as recipe adjustments, weather reports, scientific experiments, storage conditions, and energy efficiency.
II. REFERENCES and MATERIALS	Books Muschla, Judith A, et. al., 2011. <i>The Algebra Teacher’s Guide to Reteaching Essential Concepts and Skills</i> Jossey-Bass A Wiley Imprint 989 Market Street, San Francisco, CA 94103-1741 Smith, J. R. (2019). <i>Understanding Currency conversion and Global exchange Markets</i>. Financial Press. Wildi, T. (1995). Metric units and conversion charts: A metrication handbook for engineers, technologists, and scientists. IEEE Press.https://www.amazon.com/Metric-Units-Conversion-Charts-Theodore/dp/0780310500 Government Publications Macasieb, F. B., & Toledo, A. M. (2016). <i>Grade 6 Math Learner’s Module</i>. Department of Education – Bureau of Learning Resources. Retrieved from https://lrmds.deped.gov.ph/ Web References Math is Fun. (n.d.). <i>Surface Area of 3D Shapes</i>. Retrieved May 7, 2025, from https://www.mathsisfun.com/geometry/surface-area.html Khan Academy. (n.d.). <i>Surface Area</i>. Retrieved May 7, 2025, from https://www.khanacademy.org/math/geometry/hs-geo-solids CK-12 Foundation. (n.d.). <i>Surface Area of Solids</i>. Retrieved May 7, 2025, from https://www.ck12.org/geometry/surface-area/ Video lessons DepEd TV. (2021, February 12). <i>Mathematics 6 – Surface Area of Solid Figures</i> [Video]. YouTube. https://www.youtube.com/watch?v=3LtA3Xz8RGo Khan Academy. (n.d.). <i>Surface Area of Prisms, Cylinders, Cones, and Spheres</i> [Video playlist]. YouTube. https://www.youtube.com/playlist?list=PLSQ10a2vh4HCZHJ7PGeJjZlEf82m7XB-f Math Antics. (2017, August 3). <i>Surface Area</i> [Video]. YouTube. https://www.youtube.com/watch?v=7nZQzHfL6CU Materials • Calculators

	- Visual aids (3D shapes) or posters of 3D shapes/PowerPoint presentation - Worksheets (printed or digital) - Task cards/ Station Cards - Rulers, tape measures, and grid papers - Manila Papers or Cartolina - LCD Projector or TV (if using video lessons) - Application Software/ Digital Tools 1. GeoGebra 2. Desmos 3. ClassPoint																	
Lesson 4.1																		
III. CONTENT	Quantities of Different Magnitudes: Choosing Appropriate Units and Applying Scientific Notation (Recommended time: 2 hours)																	
IV. OBJECTIVES	At the end of the lesson, learners should be able to: 1. identify the appropriate metric units for measuring small and large quantities 2. apply scientific notations to express small or large numbers and vice versa. 3. solve real-life problems by applying the appropriate metric units for small and large quantities and scientific measurements.																	
V. PROCEDURES	ACTIVITIES (Instructions for Learners)	ANNOTATION (Instructions for Teachers)																
A. Activating Prior Knowledge	A.1. Activating Prior Knowledge Activity A.1 Option 1: Explore and Guess It Right! Instructions Identify the appropriate metric units for measuring small and large quantities listed below within the designated area (classroom or outdoor spaces). <table><tr><th>Quantities</th><th>Appropriate Unit of Measure</th></tr><tr><td>Station: Stretch</td><td></td></tr><tr><td>Height of the building</td><td></td></tr><tr><td>Width of the classroom</td><td></td></tr><tr><td>Length of a pencil</td><td></td></tr><tr><td>Distance from Manila to Albay</td><td></td></tr><tr><td>Station: Chill</td><td></td></tr><tr><td>Body temperature of a classmate</td><td></td></tr></table>	Quantities	Appropriate Unit of Measure	Station: Stretch		Height of the building		Width of the classroom		Length of a pencil		Distance from Manila to Albay		Station: Chill		Body temperature of a classmate		This activity refreshes measurements and conversion essential to everyday life. To facilitate this activity the teacher may apply one of the following strategies. Keep in mind that you may contextualize the item to measure depending on the interest and skills of the learners. Strategy 1: Explore and Guess It Right! 1. Provide the following materials: printed Explore and Guess It Right Activity Sheet, pencil/pen and flashcards containing the pictures of the listed quantities and different units of measure (Option 2 only) for each respective station. 2. Extra sheet of paper if preferred. 3. Explain the objective by saying “Before we learn how to apply appropriate measurements to various contexts, let’s refresh some math skills we’ll need. These skills will help us work more easily with metric conversion
Quantities	Appropriate Unit of Measure																	
Station: Stretch																		
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Width of the classroom																		
Length of a pencil																		
Distance from Manila to Albay																		
Station: Chill																		
Body temperature of a classmate																		

Temperature of Hot Coffee	
Station: Pour	
Volume of water in a pail	
Volume of a liquid in a feeding bottle	
Volume of water in a tank	
Station: Balance	
Weight of a book	
Weight of a needle	
Weight of a cow	
Station: Tick	
Time it takes to walk down the hallway	
Time it takes to finish the Scavenger Hunt	
Time it takes to travel from Sagada to Manila	

Option 2: Fastest Hunter

Instructions

Match each quantity with its corresponding appropriate unit of measure/ Write them on your paper.

Quantities	Appropriate Unit of Measure
Station: Stretch	
Height of the building	Kilometer
Width of the classroom	Meter
Length of a pencil	Centimeter
Distance from Manila to Albay	Millimeter
Station: Chill	
Body temperature of a classmate	Degree Celsius
Temperature of Hot Coffee	Degree Celsius
Station: Pour	
Volume of water in a pail	Milliliter
Volume of a liquid in a feeding bottle	Liter

focusing on length, weight, volume, temperature, and time.”

4. Present the instructions to the learners:
5. Divide the class into five (5) groups.
6. There are five measurement stations located inside the classroom—Stretch, Balance, Pour, Chill, and Tick.
7. Arrange the stations in different areas of the classroom or outdoor spaces to encourage movement and engagement ensuring the safety of the learners.
8. Learners rotate in groups from one station to the next.
9. Each group has 1 minute per station to complete the task.
10. Facilitate the discussion by briefly reviewing why using the appropriate unit of measurement is important for the lesson.

Strategy 2: Game-Based Strategy: Fastest Hunter

1. Provide the following materials:
2. Printed Explore and Guess It Right Activity Sheet., pencil/pen and flashcards containing the pictures of the listed quantities and units of measure (for option 2 only).
3. Extra sheet of paper if preferred.
4. Explain the objective by saying “Before we learn how to apply appropriate measurements to various contexts, let’s refresh some math skills we’ll need. These skills will help us work with metric conversion focusing on length, weight, volume, temperature, and time more easily.”
5. Present the instructions to the learners:
6. Divide the class into small teams.
7. There are five measurement stations located inside the classroom—Stretch, Balance, Pour, Chill, and Tick.
8. Arrange the stations in different areas of the classroom or outdoor spaces to encourage movement and engagement ensuring the safety of the learners.
9. Teams go around the stations, write their answers on the activity sheet, and submit it. The team who first correctly answers all questions in each station wins.

	<table><tr><td>Volume of water in a tank</td><td>cubic meter</td></tr><tr><td>Station: Balance</td><td></td></tr><tr><td>Weight of a book</td><td>Milligram</td></tr><tr><td>Weight of a needle</td><td>Gram</td></tr><tr><td>Weight of a cow</td><td>Kilogram</td></tr><tr><td>Station: Tick</td><td></td></tr><tr><td>Time to walk down the hallway</td><td>Hour</td></tr><tr><td>Time to finish the Scavenger Hunt</td><td>Second</td></tr><tr><td>Time it takes to travel from Sagada to Manila</td><td>Minute</td></tr></table> Answer Key: (the same for option 1 and 2) Height of the building - Meter (m) Width of the classroom - Meter (m) Length of a pencil - Centimeter (cm) Distance from Manila to Albay - Kilometer (km) Body temperature of a classmate - Degree Celsius (°C) Temperature of hot coffee - Degree Celsius (°C) Volume of water Volume of water in a pail - Liter (L) Volume of a liquid in a feeding bottle - Milliliter (mL) Volume of water in a tank - Cubic meter (m³) Weight of a book - Gram (g) or Kilogram (kg) Weight of a needle - milligram (mg) or gram (g) Weight of a cow - Kilogram (kg) Time to walk down the hallway - Second (s) Time to finish the Scavenger Hunt - Minute (min) Time it takes to travel from Sagada to Manila - Hour (h)	Volume of water in a tank	cubic meter	Station: Balance		Weight of a book	Milligram	Weight of a needle	Gram	Weight of a cow	Kilogram	Station: Tick		Time to walk down the hallway	Hour	Time to finish the Scavenger Hunt	Second	Time it takes to travel from Sagada to Manila	Minute	10. Facilitate the discussion by briefly reviewing why using the appropriate unit of measurement is important for the lesson.
Volume of water in a tank	cubic meter																			
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Time it takes to travel from Sagada to Manila	Minute																			
A.2. Establishing the Purpose of the Lesson Activity A.2: Metric Match-Up! Instructions Analyze the real-life scenarios listed below. Determine if the most appropriate metric unit is used for the measurement. Be prepared to justify your answer. 1. The most appropriate unit to measure the length of a pencil is millimeters	Activity A.2: Metric Match-Up! To facilitate this activity the teacher may apply one of the following strategies. Strategy 1: Group Discussion Instructions 1. Divide the class into small groups and assign each group a set of real-life scenarios. 2. Instruct each group to:																			

2. The most appropriate unit to measure the mass of a book is milligrams.
3. The most appropriate unit to measure the mass of a watermelon is kilograms.
4. The most appropriate unit to measure the volume of a swimming pool is cubic meters.
5. The most appropriate unit to measure the volume of a bottle of water is milliliters.
6. The most appropriate unit to measure the boiling point of water is degrees Celsius.
7. The most appropriate unit to measure the temperature of a cup of coffee is degrees Celsius.
8. The most appropriate unit to measure the duration of a movie is minutes.
9. The most appropriate unit to measure the time it takes for a runner to complete a short race is seconds.
10. The most appropriate unit to measure the height of Mayon Volcano is meters.

Answer: (Justification may vary)

1. Inappropriate. Centimeters (cm) are more suitable for everyday measurements of objects like pencils.
2. Inappropriate. Grams (g) or kilograms (kg) are more appropriate for the mass of a book.
3. Appropriate. Kilograms (kg) are the standard unit for measuring the mass of large objects.
4. Appropriate. Cubic meters (m^3) are suitable for measuring large volumes like that of a swimming pool.
5. Appropriate. Milliliters (mL) are commonly used for measuring the volume of beverages.
6. Appropriate. Degrees Celsius ($^{\circ}\text{C}$) is the metric unit for measuring temperature
7. Appropriate. Degree Celsius ($^{\circ}\text{C}$)
8. Appropriate. Minutes (min) are the standard unit for measuring time durations of this length.
9. Appropriate. Seconds (s) are the standard unit for measuring short time intervals.

Identify the most appropriate metric unit for each scenario.

Justify their choices based on the size and nature of the quantity.

Consider alternative units and explain why they are less suitable.

After discussions, each group presents their findings to the class, fostering a collaborative learning environment.

4. Connect to the lesson by stating: “In everyday life we encounter small and large quantities. Today, we’ll learn how to apply the appropriate metric units for these small and large quantities and scientific measurements.”

Strategy 2: Thumbs Up Poll

Instructions

1. Present each scenario to the class
2. Learners will show

👍 = Yes 👎 = No 🤔 = Not sure

3. Briefly ask for 1–2 volunteers to explain their answer.
4. After collecting responses, discuss the correct answers and the rationale behind them.
5. Connect to the lesson by stating: “In everyday life we encounter small and large quantities. Today, we’ll learn how to apply the appropriate metric units for these small and large quantities and scientific measurements.”

Strategy 3: Digital Poll

1. Select a digital polling tool that suits your classroom setup and objectives like Kahoot!, Slido or Google Forms.
2. Create a series of questions based on real-life scenarios
3. Present each scenario to the class, allowing learners time to read and consider their responses.
4. Launch the poll and instruct learners to submit their answers using their devices and monitor participation to ensure all learners are engaged

	10. Appropriate. Meters (m) are the standard unit for measuring heights of large structures.	5. After collecting responses, display the results to the class and discuss the correct answers and the rationale behind them, addressing any misconceptions. 6. Encourage learners to ask questions and provide explanations for their choices 7. Connect to the lesson by stating: “In everyday life, we encounter both small and large quantities. Today, we’ll learn how to apply the appropriate metric units for these quantities and scientific measurements.” 8. (Present the CS, PS, LC, Objectives, and the Unit performance Tasks)
B. Instituting New Knowledge	B.1. Presenting Examples Activity B.1: Appropriate or Not! Instructions - Your teacher will show and explain real-life examples on identifying the appropriate unit of measure - Look at each example and determine the size of the object (very small, small, medium or large sized) the type of units <i>Example 1.</i> You are tasked with hanging a large, framed picture in your classroom. To ensure it is positioned correctly, you need to measure the height from the floor to the top edge of the frame. What metric unit is best to use? <i>Answer: centimeters</i> <i>Example 2.</i> Anna is participating in a coin collection project for her mathematics class. Her task is to measure the diameter of various Philippine coins to create a comparative chart. One of the coins she will be measuring is the Philippine one-peso coin. What appropriate metric unit will she use? <i>Answer: millimeter</i>	Activity B.1: Appropriate or Not! In this activity learners will firm-up their learnings by answering another set of problems. The teacher may apply any of the following strategies. Strategy 1. Think-Pair--Share Teacher Instructions: - Give each student the learning task. - Ask them to answer each item (Think). - Then, pair them with a seatmate to discuss the items (Pair). - Finally, invite volunteers or randomly chosen pairs to share their answers and explain their reasoning to the class (Share). - You may use visual aids like posters or digital slides to show the learning task. Strategy 2. Think-Pair-Solve-Share Teacher Instructions: - Divide the class into small groups of 3-4 members. - Ask them to answer each item. - Then, discuss the answers within the group - Finally, invite volunteers or randomly chosen pairs to share their answers and explain their reasoning to the class (Share).

	B.2. Discussing New Concept Activity B.2: Option 1. Understanding the Size of SARS-CoV-2 In a local barangay hall in San Fabian, Ilocos, a health officer addresses a group of residents about COVID-19. Holding up a human hair, she explains, "This hair is about 100 micrometers in diameter." She then introduces the SARS-CoV-2 virus, noting, "The virus ranges from 60 to 120 nanometers in diameter. To put that in perspective, 1 nanometer equals 0.000000001 meters, so the virus's diameter spans from: 60 nm = 0.00000006 meters 120 nm = 0.00000012 meters Comparing the sizes, she emphasizes that the SARS-CoV-2 virus is approximately 1/1,000th the width of a human hair. This comparison underscores the virus's minuscule size and highlights the challenges in detecting and studying such a small pathogen. Understanding the scale of SARS-CoV-2 helps us appreciate the importance of preventive measures like wearing masks and practicing good hygiene, as these microscopic particles can easily spread through the air. <i>Guide Questions</i> 1. What is the approximate diameter of a human hair? 2. How does the size of SARS-CoV-2 compare to a human hair? 3. Why is it challenging to detect such a small virus without specialized equipment? 4. How can we express such tiny measurements more conveniently? Activity B.2 Option 2. Give the text “The Size of SARS-CoV-2 and its Implications by Benedette Cuffari, M.Sc”	5. You may use visual aids like posters or digital slides to show the learning task. To facilitate this activity the teacher may apply one of the following strategies. Strategy 1: Group Discussion <i>Instructions</i> 1. Divide the learners into small groups of 3–4 members. 2. Give the task and guide questions. 3. Encourage exploration. 4. After the discussion, have groups reflect on their findings and prepare to share them with the class. 5. Invite each group to present their insights and conclusions, fostering a class-wide discussion. Strategy 2: Think-Pair-Share <i>Instructions</i> Think 1. Pose the question: "How small do you think the COVID-19 virus is compared to everyday objects?" 2. Allow 2–3 minutes for learners to reflect individually. 3. Pair 4. Have learners pair up to discuss their thoughts and come to a consensus on the virus's size. Share 1. Invite pairs to share their conclusions with the class. 2. Facilitate a discussion, highlighting key insights and correcting any misconceptions. <i>Note. The teacher will discuss the key terms, definitions and examples after the activity.</i>
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https://www.news-medical.net/health/The-Size-of-SARS-CoV-2-Compared-to-OtherThings.aspx?fbclid=IwY2xjawKlwCFleHRuA2FlbQlxMABicmlkETFKY0Z5WGY4QVE2bVpMNXF2AR77H_8NngVmv8T83nnG5Sgc0ByuUDC-HgO_J4AVkiHgxrFKxxH0qdNFGqxNVg_aem_mcrwS4Axe_glQ0jHE4p8PA

Guide Questions

1. What is the approximate diameter of a human hair?
2. How does the size of SARS-CoV-2 compare to a human hair?
3. Why is it challenging to detect such a small virus without specialized equipment?
4. How can we express such tiny measurements more conveniently?
5. Reflecting on the comparisons we made, what does this activity teach us about the importance of precise and appropriate units in science and daily life?

Key Terms and Definitions

Scientific notation is a mathematical method used to express very large or very small numbers in a more concise and manageable form. It is especially useful in scientific disciplines where such numbers are prevalent. It is being represented as:

$$m \times 10^n$$

where: m is a coefficient (also called the significand or mantissa) that is greater than or equal to 1 but less than 10 and n is an integer exponent indicating the power of 10 by which the coefficient is multiplied. This notation simplifies calculations and comparisons, especially when dealing with extremely large or small quantities.

To convert a number to scientific notation:

1. Move the decimal point to create a number between 1 and 10.
2. Count the number of places the decimal was moved; this becomes the exponent of 10.
3. If the decimal was moved to the left, the exponent is positive; if to the right, it's negative.

Applying Scientific Notation to SARS-CoV-2 using scientific notation, let's consider the size of the SARS-CoV-2 virus:

Diameter: Approximately 60 to 120 nanometers (nm).

1 nanometer (nm) = 1×10^{-9} meters.

Converting to meters:

60 nm = 0.00000006 meters = 6×10^{-8} meters.

120 nm = 0.00000012 meters = 1.2×10^{-7} meters.

Expressing these measurements in scientific notation makes it easier to comprehend and compare the minuscule size of the virus. It is also more convenient for comparisons and mathematical operations, like multiplication and division.

Standard notation is the conventional way of writing numbers using digits, without exponents or powers of ten. It's the format we commonly use in everyday life, such as 4,500,000,000 or 0.00000036. In standard notation, each digit has a specific place value: units, tens, hundreds, etc., which influences the overall value of the number. This format is intuitive and straightforward for representing numbers in their usual form.

To convert scientific notation to standard notation:

1. Identify the exponent.
2. Move the decimal point in the base number according to the exponent. (If the exponent is positive, move the decimal point to the right; if negative, to the left.)
3. Write the result.

For example, the average distance from Earth to the Sun is approximately 1.496×10^8 kilometers. Express the average distance to standard form.

Solution

1. Identify the Exponent:

	2. The exponent is 8, indicating the number of places to move the decimal point. 3. Move the Decimal Point: 4. Starting with 1.496, move the decimal point 8 places to the right: 1.496 → 14.96 → 149.6 → 1496 → 14,960 → 149,600 → 1,496,000 → 14,960,000 → 149,600,000 5. Result: The standard notation is 149,600,000 kilometers, representing the distance from Earth to the Su	
	B.3. Developing Mastery Activity B.3: Measure It Right: Math in Everyday Life Instructions Complete the given tasks. Part 1: Identify the Appropriate Metric Unit 1. measuring the length of a football field 2. weighing a newborn baby 3. determining the volume of a soft drink can 4. calculating the distance between two cities 5. timing a sprint race 6. measuring the height of a tree 7. measuring the temperature of a patient. 8. measuring the mass of a chemical compound Part 2: Convert to Scientific Notation Express each of the following numbers in scientific notation: 1. 0.000451 2. 78,000 3. 56,000,000 4. 123,000 5. 0.00000032 6. 0.0000071 7. 0.0000000000053 8. 0.00000000379 Part 3: Convert to Standard Notation Rewrite each of the following numbers in standard form: 1. 4.5×10^{-4} 5. 9.3×10^{-5}	Activity B.3: Measure It Right: Math in Everyday Life In this activity learners will firm-up their learnings by answering another set of problems. The teacher may apply any of the following strategies. Strategy 1. Think-Pair-Solve-Share Instructions: 1. Give each student the learning task. 2. Ask them to answer/solve one item (Think). 3. Then, pair them with a seatmate to discuss and solve the next one together (Pair + Solve). 4. Finally, invite volunteers or randomly chosen pairs to share their answers and explain their reasoning to the class (Share). 5. You may use visual aids like posters or digital slides to show the learning task. Strategy 2. Game-Based: Metric & Scientific Notation Challenge Instructions 1. Prepare materials: Measurement Cards, Number Cards, tokens/markers, and Baamboozle (or similar game). 2. Divide the class into groups of 3–4. 3. A team draws a Measurement Card → identify the correct unit. 4. The same team draws a Number Card → convert to scientific notation.

	2. 3.2×10^6 3. 8.1×10^{-7} 4. 6.7×10^3 6. 5.6×10^4 7. 1.2×10^{-2} 8. 7.8×10^8 Part 4: Word Problems Use a scientific calculator or a computing device to solve the following word problems involving scientific notation. write your answer in scientific notation. - Light travels at a speed of 3.00×10^8 meters per second. The distance from Earth to the Sun is approximately 1.496×10^8 kilometers. How many seconds does it take for light to travel from the Sun to Earth? Express your answer in scientific notation. - The diameter of the Sun is approximately 1.39×10^6 kilometers, while the diameter of Earth is about 1.27×10^4 kilometers. How many times larger is the diameter of the Sun compared to Earth? Express your answer in scientific notation. - The world population in 2025 is estimated to be 8.5 billion people. The population of a city is 4.2×10^6 people. What is the ratio of the city's population to the world population? Express your answer in scientific notation. - The Alpha Centauri system is approximately 4.37 light-years away from Earth. A spacecraft travels at a speed of 2.5×10^4 kilometers per second. If the spacecraft could travel directly to the Alpha Centauri system without stopping, how many seconds would it take to reach it? Express your answer in scientific notation. Note: 1 light-year $\approx 9.461 \times 10^{12}$ kilometers.	- The team shares answers; teacher/other teams check correctness. - Give points/tokens for correct answers. - Repeat turns until all teams have played. - Use Baamboozle on the screen for extra interactive rounds. <i>Answer Key</i> Part 1 - meter - kilogram - milliliter - kilometer - second - Meter - Degree Celsius - Gram Part 2 4.51×10^{-4} 7.8×10^4 5.6×10^7 1.23×10^5 3.2×10^{-7} 7.1×10^{-6} 5.3×10^{-11} 3.79×10^{-9} Part 3 0.00045 3,200,000 0.00000081 6,700 0.000093 56,000 0.012 780,000,000
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		Part 4 1. 4.99×10^2 seconds 2. 1.09×10^2 times larger 3. 4.94×10^{-4} 4. 1.65×10^9 seconds
C. Demonstrating Knowledge and Skills	C.1. Finding Practical Application Activity C.1: Practical Applications of Scientific Measurement Instructions: Read and analyze each scenario and answer the questions. <i>Practical Application 1.</i> You are a laboratory technician tasked with preparing a saline solution for an experiment. The recipe specifies that the solution should contain 0.9 grams of sodium chloride (NaCl) dissolved in 100 milliliters of water. Are the metric units for measuring the weight of sodium chloride and volume of water appropriate? Justify your answer. <i>Answer.</i> Yes, the metric units used are appropriate. These units ensure accuracy and consistency in preparing the saline solution. Practical Application 2. As a medical laboratory technician in a hospital in Metro Manila, you are analyzing a blood sample. The average diameter of a human red blood cell is approximately 0.000007 meters. To facilitate documentation and analysis, you need to express this measurement in scientific notation. Which of the following is the correct scientific notation for the average diameter of a human red blood cell? Explain your answer. A. 7×10^{-5} meters B. 7×10^{-6} meters C. 7×10^{-7} meters D. 7×10^{-8} meters	Activity C.1: Practical Applications of Scientific Measurement To facilitate this activity the teacher may apply one of the following strategies. Strategy 1. Think-Pair-Share Learners will: Think: Reflect individually on the given problem Pair: Discuss their thoughts with a partner. Share: Present their ideas to the class. Strategy 2. Group Activity Divide the class into a small group and give the task. Ensure that all learners actively participate. Provide timely and constructive feedback to support their learning process

	Answer: B. 7×10^{-6} meters. This is achieved by moving the decimal point 6 places to the right, transforming the number into 7, and indicating the movement with a negative exponent, reflecting the small size of the measurement	
	C.2. Making Generalization Activity C.2 Option 1. I Learned, I Wonder <i>Instructions:</i> 1. Fold a sheet of paper in half. 2. On one side, write: "I Learned..." – list 3 things you now understand about: 3. Identifying appropriate metric units for measuring small and large quantities. 4. Applying scientific notation to express small or large numbers. 5. Solving real-life problems by applying the appropriate metric units and scientific measurements. 6. On the other side, write: "I Wonder..." – pose 2–3 questions or reflections you still have. 7. Volunteers will share their responses. Option 2. Concept Map <i>Instructions:</i> 1. Draw a central circle labeled "Measurement and Scientific Notation". 2. Build branches with related concepts such as: 3. Metric Units: millimeter (mm), centimeter (cm), meter (m), kilometer (km), milligram (mg), gram (g), kilogram (kg), microgram (μg), nanometer (nm). 4. Scientific Notation: expressing numbers like 0.00000036 as 3.6×10^{-7}. 5. Real-Life Applications: measuring the size of a red blood cell (7×10^{-6} m), the distance from Earth to the Sun (1.496×10^8 km), or the mass of a virus (2.5×10^{-12} kg).	To wrap-up the lesson, you may apply one of the following strategies the provided activities.

	6. You will work in groups or as a class to fill it in.							
	C.3. Evaluating Learning Activity C. 3: My Learning Proficiency <i>Instructions:</i> Read and answer the following completely. 1. What metric unit would you use to measure the mass of a grain of rice? Justify your answer. 2. The diameter of a human hair is approximately 8×10^{-5} meters, and the diameter of a red blood cell is about 7×10^{-6} meters. Which is smaller, and in what ratio? 3. You are a microbiologist conducting research on bacterial growth. During your study, you measure the length of a bacterium and find it to be 0.0000025 meters. To accurately record and communicate this measurement, you decide to express it in scientific notation. What is the length of the bacterium in scientific notation? Explain why using scientific notation is beneficial when dealing with measurements of microorganisms. 4. In scientific research, why is it important to express measurements like the speed of light (approximately 299,792,458 meters per second) in scientific notation?	Activity C. 3: My Learning Proficiency Each student is required to complete this activity independently to demonstrate their understanding of the lesson. Timely feedback will be provided to support learners who may need additional assistance or clarification. <i>Answers:</i> 1. The appropriate metric unit to measure the mass of a grain of rice is the milligram (mg) because a single grain of rice typically weighs around 25 milligrams (0.025 grams), which is a very small mass. 2. A red blood cell is smaller in diameter than a human hair. The ratio of the diameter of a red blood cell to that of a human hair is 0.875, i.e., the diameter of a human hair is 11.4 times that of a red blood cell. 3. The length of the bacterium in scientific notation is 2.5×10^{-6} meters. Scientific notation provides a concise and standardized way to express measurements of microorganisms, whose sizes are often on the microscopic scale. 4. Scientific notation provides ease of calculation for handling very large or very small measurements.						
	C.4. Additional Activities Activity C.4: Reinforcement Tasks <i>Remediation Activity</i> A. Match each item with the most appropriate metric unit. Write the correct letter in the space provided. <table><tr><td>Item</td><td>Metric Unit</td></tr><tr><td>Mass of a grain of rice</td><td>A. meter (m)</td></tr><tr><td>Length of a classroom</td><td>B. milligram (mg)</td></tr></table>	Item	Metric Unit	Mass of a grain of rice	A. meter (m)	Length of a classroom	B. milligram (mg)	Activity C.4: Reinforcement Tasks This exercise is designed for individual completion to assess each learner's understanding of the lesson. Based on their performance, differentiated support will be provided:
Item	Metric Unit							
Mass of a grain of rice	A. meter (m)							
Length of a classroom	B. milligram (mg)							

	Distance from Earth to the Moon C. kilometer (km) Volume of a swimming pool D. cubic meters (m³) Mass of a paperclip E. gram (g) Reflection: Explain why you chose each metric unit for the corresponding item. B. Convert the following numbers as indicated. 1. 0.00045 → Scientific Notation: 2. 700,000,000,000 → Scientific Notation: 3. 4×10^7 → Standard Form: 4. 1.53×10^{-4} → Standard Form: Hint: To convert to scientific notation, move the decimal point so that only one non-zero digit remains to its left. Count the number of places moved; this becomes the exponent of 10. C. Answer the following real-life problems 1. The mass of a grain of rice is approximately 0.000025 kilograms. Express the mass in scientific notation 2. Number of Cells in the Human Body is approximately 3.72×10^{13}. Express it in standard form. Enrichment: Real-World Scenario Project Instructions: 1. Pick a topic with measurements (pollution, space, disasters, health, etc.) 2. Collect recent and reliable data 3. Convert all data to metric units 4. Write large/small numbers in scientific notation 5. Analyze the data for patterns or trends 6. Write a report: a. Introduction b. Methods	Remediation: For learners who have not yet achieved mastery, targeted interventions will be implemented to reinforce foundational concepts and skills. Enrichment: For learners who have demonstrated proficiency, extension activities will be offered to deepen their understanding and challenge their thinking.
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	c. Results d. Discussion e. Conclusion f. References 7. Prepare a 5–7 min presentation	
Lesson 4.2		
III. CONTENT	Unit Conversion Within and Across Systems <i>(Recommended time: 2 hours)</i>	
IV. OBJECTIVES	At the end of the lesson, learners should be able to: 1. differentiate the metric and English (imperial) systems of measurement, including their units for length, mass, volume, time and temperature. 2. convert measurements between different units within the metric and within English system 3. convert measurements from the metric system to the English system and vice versa 4. solve practical problems involving unit conversions, such as cooking recipes, travel distances, and scientific experiments.	
V. PROCEDURES	ACTIVITIES	ANNOTATION
A. Activating Prior Knowledge	A.1. Activating Prior Knowledge Activity A.1: Exploring Measurement Systems Option 1: Pinoy Travel Challenge Read and analyze the scenario below and answer the guide questions. <i>Scenario</i> You have won an international travel package to the United States. Before your trip, you need to check your luggage weight and the weather forecast to pack appropriately. Some measurements are in kilograms and Celsius, others in pounds and Fahrenheit. <i>Guide Questions</i> Your suitcase weighs 23 kilograms. The airline baggage allowance is 50 pounds. Is your suitcase within the airline limit? Explain your answer.	Activity A.1 Exploring Measurement Systems These activities will help learners understand why knowing how to convert between metric and English systems is essential in real life, especially when traveling internationally. They will apply conversion skills to luggage weight and temperature to make smart decisions. You may facilitate any of the provided activities.

	The temperature forecast for your travel dates shows 86°F. What type of clothes should you pack? Explain. Why is it important to understand different units like pounds and Fahrenheit when traveling internationally? Option 2: Metric or English? Let’s Race! <i>Instructions:</i> Work with your team to guess the right unit system, convert measurements, and solve a real-life problem — all in 5 minutes! Round 1: Metric or English? (one point each for 10 seconds) <table><tr><td>feet →</td><td>English</td></tr><tr><td>kilometer →</td><td>Metric</td></tr><tr><td>centimeter →</td><td>Metric</td></tr><tr><td>miles →</td><td>English</td></tr><tr><td>meters →</td><td>Metric</td></tr></table> Round 2: Convert It! (two points each for 1 minute) Convert 1 foot to inches Convert 1 meter to centimeters Convert 3 minutes to seconds Round 3: Real-Life Scenario (one point for 1 minute) A recipe says 2 cups of milk, but you only know milliliters. What should you do?	feet →	English	kilometer →	Metric	centimeter →	Metric	miles →	English	meters →	Metric	
feet →	English											
kilometer →	Metric											
centimeter →	Metric											
miles →	English											
meters →	Metric											
	A.2. Establishing the Purpose of the Lesson Activity A.2: When Units Go Wrong Option 1: Crash and Learn “A student mixed up inches and centimeters when building a birdhouse and it turned out way too small.” What went wrong? How could this have been prevented? Option 2: Would You Rather? “Would you rather run 5 kilometers or 3 miles? Would you rather have 1 liter or 4 cups of ice cream?” How do you know which one is more? Why might it be important to convert units before deciding?	Activity A.2: When Units Go Wrong The provided activities will engage learners in comparing measurements in different units. It highlights the need to convert before making decisions and builds awareness of how unit systems affect our choices. Choose one these activities and facilitate applying any of the following strategies. Strategy 1: Reflective Practice 1. Present the Scenario (Whole Class or Small Group) 2. Allow group discussion. Questions are as follows: 3. What is the difference between inches and centimeters?										

		4. Where might the student have made the mistake? 5. Why does this matter? 6. Encourage learners to share real-life moments when a small mistake caused a big problem. Strategy 2: Think-Pair-Share 1. Think quietly and write down your answer. 2. Turn to a partner. Share your answer and explain your thinking. Listen to the answer. Do you agree or disagree? 3. Ask volunteers to share their ideas with the class.
B. Instituting New Knowledge	B.1. Presenting Examples Activity B.2: Measurements in Daily Activities Instructions: Read and try to solve each problem. <i>Problem 1: Running a Race</i> You are training for a race. The race is 5 kilometers long, but you usually think in miles. You want to know how far the race is in miles. <i>Questions</i> How many miles is the race? Why would it be helpful to know the distance in miles before you run? <i>Problem 2: Making a Smoothie</i> You want to make a smoothie. The recipe says you need 1 liter of milk, but your measuring cup shows cups not liters. You need to figure out how many cups there are in 1 liter. <i>Questions:</i> How many cups of milk do you need for the smoothie? Why is it important to convert liters to cups when following a recipe?	Activity B.2: Measurements in Daily Activities The activity introduces unit conversions by using practical scenarios that are easy for learners to relate to. They demonstrate how converting between different units is important in daily activities like sports and cooking. This will lead into the lesson on conversion formulas, showing learners how to apply these conversions in various contexts. To facilitate this activity, follow these steps: 1. Present the scenario 2. Ask students to answer the questions. 3. Guide them to reflect. 4. Facilitate a short discussion linking conversion to making realistic plans and goals.
	B.2. Discussing New Concept	Activity B.2: Understanding and Applying Unit Conversion

Activity B.2:

Understanding and Applying Unit Conversion

Unit conversion is the process of changing one unit of measurement to another. It's important because different systems use different units, and we often need to convert between them.

Within the Same System (e.g., Metric System):

In the metric system, we convert by multiplying or dividing by powers of 10.

Example: 1 kilometer = 1,000 meters.

To convert 3 kilometers to meters:

$$3 \text{ km} \times \frac{1000 \text{ m}}{1 \text{ km}} = 3,000 \text{ meters.}$$

Across Systems (e.g., Metric to English):

To convert between different systems, we use conversion factors.

Example: 1 inch = 2.54 centimeters.

To convert 5 inches to centimeters:

$$5 \text{ inches} \times \frac{2.54 \text{ cm}}{1 \text{ in}} = 12.7 \text{ centimeters.}$$

Steps for Conversion:

Identify the units.

Find the conversion factor.

Multiply or divide by the factor.

Write the result with the correct units.

Problem 1: You are training for a race. The race is 5 kilometers long, but you usually think in miles. You want to know how far the race is in miles. How many miles is the race?

Solution:

To convert kilometers to miles, you use the conversion factor:

$$1 \text{ mile} \approx 1,609 \text{ kilometers}$$

So, for a 5-kilometer race:

This part is crucial, appropriate strategy is essential to be for the learners to understand and apply the skills. To facilitate this, you may follow the suggested procedures below.

1. Introduce the Concept

- Begin by asking students: “Have you ever needed to change a measurement, like kilometers to meters or cups to liters, to understand it better?”
- Explain: “This process is called **unit conversion**—changing one unit into another so measurements make sense in different situations.”
- Present the common conversion factors.

Common Unit Conversion Factors

Length

Within	Metric	1 centimeter=10 millimeters 1 meter = 100 centimeters 1 kilometer = 1,000 meters
	English	1 foot = 12 inches 1 yard = 3 feet 1 mile = 5,280 feet
Across		1 inch $\approx$ 2.54 centimeters 1 foot $\approx$ 0.3048 meters 1 mile $\approx$ 1.609 kilometers

Mass / Weight

Within	Metric	1 gram=1000 milligrams 1 kilogram = 1,000 gram 1 metric ton=1000 kilograms
	English	1 pound = 16 ounces 1 ton (U.S.) = 2,000 pounds
Across		1 ounce $\approx$ 28.35 grams 1 pound $\approx$ 0.4536 kilograms 1 kilogram $\approx$ 2.2046 pounds

Volume / Capacity

Within	Metric	1 liter = 1,000 milliliters 1 cubic meter = 1,000 liters
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	$5 \text{ km} \times \frac{1 \text{ mile}}{1.609 \text{ km}} = 3.1075201989 \text{ miles}$ Answer: The race is approximately 3.11 miles long. Problem 2: You want to make a smoothie. The recipe says you need 1 liter of milk, but your measuring cup shows cups not liters. How many cups of milk do you need for the smoothie? Solution: To convert liters to cups, use the conversion: $1 \text{ liter} \approx 4.22675 \text{ US cups}$ So, if the recipe calls for 1 liter of milk: $1 \text{ liter} \times \frac{4.22675 \text{ cups}}{1 \text{ liter}} = 4.22675 \text{ cups}$ Answer: You need approximately 4.23 cups of milk.	<table border="1" data-bbox="1317 148 2132 403"> <tr> <td data-bbox="1317 148 1442 263"></td><td data-bbox="1442 148 1588 263">English</td><td data-bbox="1588 148 2132 263"> 1 gallon (U.S.) = 4 quarts 1 quart = 2 pints 1 pint = 2 cups </td></tr> <tr> <td colspan="2" data-bbox="1317 263 1588 403">Across</td><td data-bbox="1588 263 2132 403"> 1 ounce $\approx$ 29.57 milliliters 1 gallon (U.S.) $\approx$ 3.785 liters 1 liter $\approx$ 0.2642 gallons (U.S.) 1 liter $\approx$ 4.22675 US cups </td></tr> </table> Time (Universal Across Systems) 1 minute = 60 seconds 1 hour = 60 minutes 1 day = 24 hours 1 week = 7 days Temperature Celsius to Fahrenheit: $^{\circ}\text{F} = (^{\circ}\text{C} \times 9/5) + 32$ Fahrenheit to Celsius: $^{\circ}\text{C} = (^{\circ}\text{F} - 32) \times 5/9$ Celsius to Kelvin: $\text{K} = ^{\circ}\text{C} + 273.15$ When the Philippines switched to the metric system people were encouraged to memorize the conversion rules by playing an ad repeatedly on TV and radio. It helped many learners in the 70's learn the Imperial system to the metric system conversion. The teacher might want to share the song with the class. https://www.youtube.com/watch?v=RwQ404hewR4 • How can one use this song to convert from metric to English? 2. <i>Explain Within-System Conversions (Metric)</i> • Write the example on the board: <i>1 kilometer = 1,000 meters.</i> • Show the step: $3 \text{ km} \times 1000 = 3000 \text{ m.}$ • Emphasize that the metric system uses powers of 10, making conversion easier. 3. <i>Explain Across-System Conversions (Metric $\leftrightarrow$ English)</i> • Give the example: <i>1 inch = 2.54 centimeters.</i> • Work through: $5 \text{ in} \times 2.54 = 12.7 \text{ cm.}$		English	1 gallon (U.S.) = 4 quarts 1 quart = 2 pints 1 pint = 2 cups	Across		1 ounce $\approx$ 29.57 milliliters 1 gallon (U.S.) $\approx$ 3.785 liters 1 liter $\approx$ 0.2642 gallons (U.S.) 1 liter $\approx$ 4.22675 US cups
	English	1 gallon (U.S.) = 4 quarts 1 quart = 2 pints 1 pint = 2 cups						
Across		1 ounce $\approx$ 29.57 milliliters 1 gallon (U.S.) $\approx$ 3.785 liters 1 liter $\approx$ 0.2642 gallons (U.S.) 1 liter $\approx$ 4.22675 US cups						

		• Highlight that across systems we need conversion factors (not just moving decimals). 4. <i>Teach the 4 Steps of Conversion</i> • Display the steps: 1. Identify the units. 2. Find the conversion factor. 3. Multiply or divide. 4. Write the result with units. • Ask students to repeat the steps aloud as a memory aid. 5. <i>Work on Practice Problems (Guided Practice)</i> • Present Problem 1 (Running a Race). Solve together on the board. • Present Problem 2 (Making a Smoothie). Solve together with class participation. <i>Note: You may provide more examples if necessary.</i>
	B.3. Developing Mastery Activity B.3: The Great Conversion Escape Challenge <i>Instructions:</i> 1. Form groups of 3-5 members. 2. Each group will receive a set of Conversion Challenge Cards with different problems (length, volume, weight, temperature, and time). 3. Solve each card step by step and show your work. Example challenges include: a. Convert 8 liters to milliliters b. Convert 25°C to Fahrenheit c. A trip is 120 miles long. How many kilometers is that? d. A box of flour weighs 2.5 kilograms. How many pounds is that? e. Convert 45 minutes to seconds 4. After completing your card set, you will unlock the Escape Stations. At each station: a. Solve the challenge.	Activity B.3: The Great Conversion Escape Challenge This activity is a fun and interactive way for learners to practice converting measurements across metric and English systems. Students work in small groups to solve real-life conversion problems presented through challenge cards and an escape room mission. To facilitate the activity, follow the instructions and facilitate the following questions: 1. Which challenge was the easiest or hardest? Why? 2. How did your team divide the work? 3. Why is unit conversion important in real-life situations (travel, recipes, science experiments, etc.)?

	b. Escape Stations Station 1: The Water Lock Convert 2,500 mL to liters Answer: 2.5 L → Clue Code: 2 Station 2: Temperature Code Convert 30°C to Fahrenheit Answer: 86°F → Clue Code: 8 Station 3: Volume Vault Convert 4 quarts to cups Answer: 16 cups → Clue Code: 1 a. Record the first digit of your answer as a Clue Code. b. Use all Clue Codes to form the Final Escape Code. 5. Final Step Combine your clue digits to unlock the vault! Final Escape Code: 2 – 8 – 1	
C. Demonstrating Knowledge and Skills	C.1. Finding Practical Application Activity C.1: Real-Life Conversion Scenarios <i>Instructions:</i> Analyze the given scenarios and provide necessary solutions <i>Scenario 1: Cooking from an International Recipe</i> You find a delicious French recipe for lentil soup that calls for 1.2 liters of vegetable broth. You're in the U.S., and your measuring tools are in cups, so you do the conversion: $1.2 \text{ liters} \times \frac{4.22675 \text{ cups}}{1 \text{ liter}} = \text{about } 5.1 \text{ cups}$ Now you know to use just over 5 cups of broth. <i>Scenario 2: Dress Production</i> Maria owns a small garment workshop in Manila. She receives an order from a boutique in Australia for 200 custom summer dresses. Each dress requires 2.5 yards of soft cotton fabric based on the pattern layout. Maria	Activity C.1: Real-Life Conversion Scenarios This activity helps learners understand how measurement conversions are applied in real-world contexts such as cooking and business. By working through practical scenarios, students practice converting between metric and English units, making accurate decisions, and recognizing why correct conversions are essential in everyday life. To facilitate this activity here are the suggested strategies. 1. Divide the class into pairs or small groups. 2. Distribute the scenarios and allow learners to read carefully. 3. Ask the learners to perform the conversions. 4. Facilitate the group discussion. a. Why is it important to convert correctly in each case? b. What might go wrong if the conversion is inaccurate?

	contacts her fabric supplier in China, but the supplier provides fabric measurements and prices in meters, not yards. To avoid ordering the wrong amount, Maria needs to convert her fabric requirement from yards to meters. Fabric Conversion: Fabric needed (in yards): $200 \text{ dresses} \times 2.5 \text{ yards} = 500 \text{ yards}$ Convert to meters (1 yard = 0.9144 meters): $500 \times 0.9144 = 457.2 \text{ meters}$ Add 10% for cutting waste and shrinkage: $457.2 \times 1.10 = 502.92 \text{ meters}$ Thus, Maria orders 503 meters of fabric to complete the dresses without shortages.	5. Ask each group to prepare their output. 6. Each group presents their solutions and insights.
	C.2. Making Generalization Activity C.2: Wrapping Up: Key Takeaways on Unit Conversions Option 1. Crossing the Conversion Bridge Instructions: 1. Take a sheet of paper and divide it into two sections: "I Learned..." and "I Still Need to Learn..." 2. On the "I Learned..." side, write 3 things you understand about: 3. Converting between different units within the metric system (e.g., kilometers to meters, milliliters to liters). 4. Converting between the metric and imperial systems (e.g., kilograms to pounds, kilometers to miles). 5. The importance of using conversion factors (e.g., 1 inch = 2.54 cm, 1 liter = 1.057 quarts). 6. On the "I Still Need to Learn..." side, write 2-3 questions you have about converting between systems (e.g., "When do I use rounding in conversions?" or "What are the most common errors when converting units?"). 7. Share your thoughts with a partner or in a group discussion to see if others have similar questions or insights.	Activity C.2: Wrapping Up: Key Takeaways on Unit Conversions To wrap up the lesson, you may use one of the following strategies through the provided activities. .

	Option 2. Unit Conversion Concept Map: Across Systems Instructions: 1. Draw a large circle in the center of your paper and label it “Unit Conversion.” 2. Create branches for: • Metric Units (mm, cm, m, km, g, L). • English Units (inches, feet, miles, pounds, gallons). 3. Add sub-branches that show: • Conversions within the Metric System. • Conversions between Metric and Imperial Systems. 4. Include real-life applications such as: • Travel: kilometers ↔ miles. • Science: milliliters ↔ liters, grams ↔ kilograms. • Daily Use: cups ↔ liters, ounces ↔ grams. 5. Work in pairs or small groups to complete your concept map and compare your approaches.									
	C.3. Evaluating Learning Activity C.3: Checking What We Learned <i>Instructions:</i> Answer each question carefully. Show your calculations and include the correct units. A. You are preparing for a hiking trip to Mount Pulag, the highest peak in Luzon, Philippines, standing at 2,928 meters (9,606 feet) above sea level. Known for its breathtaking "sea of clouds" and diverse ecosystems, it's a popular destination for nature enthusiasts. Convert the following measurements to the specified units: 1. The trail you plan to hike is 15 kilometers long. Convert this distance to miles. 2. You have a 5-liter water bottle. Convert this volume to milliliters. 3. Your backpack weighs 3 kilograms. Convert this weight to pounds.	Activity C.3: Checking What We Learned <i>Instructions:</i> 1. Use this task as both individual work and an opportunity for group discussion to deepen understanding. 2. Allow learners to peer review their answers using the rubric. 3. Incorporate error analysis by showing common mistakes and asking learners to correct them. 4. Use the following to evaluate each student's work: <table><tr><th>Criteria</th><th>Indicators of Mastery</th></tr><tr><td>Correct Calculation</td><td>Uses correct math operations and conversion factors</td></tr><tr><td>Units Accuracy</td><td>Includes and labels units appropriately for all answers</td></tr><tr><td>Work Shown</td><td>Steps are clear, logical, and follow a structured format</td></tr></table>	Criteria	Indicators of Mastery	Correct Calculation	Uses correct math operations and conversion factors	Units Accuracy	Includes and labels units appropriately for all answers	Work Shown	Steps are clear, logical, and follow a structured format
Criteria	Indicators of Mastery									
Correct Calculation	Uses correct math operations and conversion factors									
Units Accuracy	Includes and labels units appropriately for all answers									
Work Shown	Steps are clear, logical, and follow a structured format									

	4. The elevation gain on the trail is 800 meters. Convert this elevation to feet. 5. You estimate the hike will take 6 hours. Convert this time to minutes. B. You are organizing a community event and need to prepare a large batch of fruit punch. The recipe requires 5 gallons of fruit juice. However, your measuring equipment uses liters and cubic centimeters. 1. Calculate the total volume of fruit juice required in cubic centimeters. 2. Determine how many standard-sized plastic bottles, each holding 1.5 liters, are needed to store the entire batch.	<table><tr><th>Real-Life Connection</th><th>Recognizes why each conversion is useful or relevant (as a bonus)</th></tr></table> 5. Score each problem out of 4 points (as per rubric). 6. Total score: 20 points 7. Use results to group learners for targeted instruction (e.g., reteach, enrich). Answer Key: A. 1. 9.2 miles 2. 5,000 mL 3. 661 lb 4. 2,625 ft 5. 360 min B. 1. 18,927 cm3 2. 13 bottles 3. 16 bottles	Real-Life Connection	Recognizes why each conversion is useful or relevant (as a bonus)
Real-Life Connection	Recognizes why each conversion is useful or relevant (as a bonus)			
	C.4. Additional Activities Activity C.4: Practice More, Learn More Remediation Activity: Unit Conversion Practice Materials: Flashcards, paper, or whiteboard <i>Instructions:</i> 1. Work with a partner. 2. Take turns asking and answering questions using the flashcards. 3. Write down or say your answer. 4. Check the back of the flashcard to see if you are correct. 5. If your answer is wrong, talk with your partner about the correct conversion. 6. Switch roles so you both get a chance to answer and ask. Enrichment Activity: Real-Scenario Conversion Materials: Paper, internet access (if needed), presentation tools	Activity C.4: Practice More, Learn More The differentiated instructional strategies address diverse learner needs in mastering unit conversions. For the Remediation Activity: 1. Develop and organize flashcards containing unit conversion problems and corresponding answers. 2. Structure student pairings to facilitate peer-assisted learning. 3. Observe and support student interactions to ensure accuracy in responses and provide immediate corrective feedback where necessary. 4. Lead a reflective class discussion to address common misconceptions and reinforce accurate understanding of unit conversions. For the Enrichment Activity: 1. Assist learners in selecting appropriate and meaningful real-world scenarios requiring unit conversions. 2. Guide learners in identifying relevant data and constructing accurate conversion charts.		

	<i>Instructions:</i> 1. Pick a real-life situation where unit conversion is needed. Examples: Calculating the fuel efficiency of a car (miles per gallon → kilometers per liter). Planning a trip (miles → kilometers, pounds → kilograms). Comparing cooking measurements (cups, teaspoons → liters, milliliters). 2. List all the conversions you need for your chosen scenario. Show the steps and answers clearly. 3. Explain your scenario, show your conversions, and describe why the conversion is important in that situation. 4. Present your work so others can learn from your example.	3. Offer ongoing feedback throughout the project to support conceptual clarity and procedural accuracy. 4. Evaluate learners' comprehension through the quality of their application, reasoning, and presentation of unit conversions
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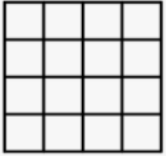
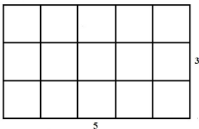
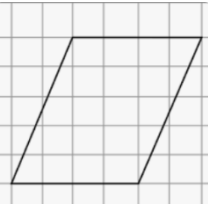
Lesson 4.3

III. CONTENT	Determining the Surface Area of Three-Dimensional Objects <i>(Recommended Time 2 Hours)</i>
IV. OBJECTIVES	At the end of the lesson, learners should be able to: 1. identify the basic 3D figures such as prisms, pyramids, cylinders, cones, and spheres and their different parts using blueprints and models. 2. select and apply the appropriate surface area formula for each type of solid. 3. compute the surface area of individual 3D objects accurately using given measurements. 4. solve real-life problems involving surface area of 3D objects or solid figures and their combinations.

IV. PROCEDURES	ACTIVITIES	ANNOTATION
A. Activating Prior Knowledge	A.1. Activating Prior Knowledge Activity A.1: Warm-Up Discussion Part 1. Area Review Using Grids <i>Instructions:</i> 1. You will be given a worksheet or digital slide with 5 shapes drawn on square grids. 2. Each shape fits on a grid where each square = 1 square unit.	Activity A.1: Warm-Up Discussion This activity is designed to activate learners' prior knowledge on 2D and 3D geometry before introducing new concepts. In Part 1, learners review the concept of area by working with shapes on grids, reinforcing both counting and formula-based solutions. In Part 2, they connect abstract geometric ideas to real-life examples by identifying common 3D objects and describing their surface types. In Part 3, learners apply observation and classification skills in a "3D Object Hunt," where they list real-world objects,

3. For each shape:
 - a. Count or calculate the area using the grid.
 - b. Write the calculated number of squares in the column “Number of Square Units”.
4. Write the formula used.
5. Show your solution (area counting or formula-based).
6. Compare your answers with a seatmate after completing.

Shapes on the Grid Worksheet:

2D Shape	Illustration	Number of Square Units	Formula for Area
Square			
Rectangle			
Triangle			
Parallelogram			

match them with geometric solids, and describe their surface characteristics. This warm-up encourages recall, discussion, and real-world connections, preparing learners for deeper exploration of surface area and volume. To facilitate this activity the teacher may apply following strategies.

Part 1. Area Review Using Grids

Instructions:

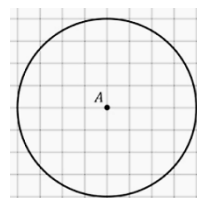
1. Prepare a worksheet or slide showing each shape on square grids.
2. Review each area formula orally before the activity begins.
3. Encourage learners to visualize the area by counting full squares and estimating partials when needed.
4. Observe learner strategies: Are they relying on formulas, counting, or both?

Part 2. 3D Object Hunt and Talk

Instructions:

1. Organize Groupings. Facilitate the formation of groups with 5 members each, ensuring a mix of abilities and encouraging collaboration.
2. Set the Purpose of the Activity. Explain the purpose of the warm-up activity, relating it to the upcoming lesson on 3D geometric figures and their surface characteristics.
3. Prompt Thinking with Guiding Questions. Ask questions such as:
 “What 3D shapes do you commonly see at home or in school?”
 “What do their surfaces look or feel like?”
4. Use concrete examples or visuals (e.g., a ball for a sphere, a box for a cube).
5. Monitor and Support Group Discussions. Circulate around the classroom, listening to group conversations, guiding them with prompts, and addressing misconceptions.

Circle



Part 2. 3D Object Hunt and Talk

Instructions:

1. Form groups of 5 members.
2. Think about the world around you: your home, classroom, or community. What are some 3D objects you see at home or in school?
3. What does the surface of these objects look like?
4. Recall and connect real-life 3D objects to their geometric names and surface characteristics.

Part 3: 3D Object Hunt

Instructions:

1. Look around the classroom or think of real-life objects you know or see at home, in stores, etc.) that match 3D shapes.
2. List at least 5 objects, along with the name of the object, the 3D shape it resembles (e.g. prism, pyramid, cylinder, cone, sphere), whether it has flat surfaces, curved surfaces or both, and estimate how many surfaces or faces it has. You may draw quick sketches of the objects next to their names.
3. Use the template for each group to fill out.

Real-Life Object	Geometric Shape	Surface Type (Flat/ Curved/ Both)	Estimated Number of Surfaces
Example: Shoe Box	Rectangular Prism	Flat only	6

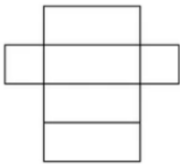
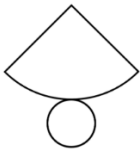
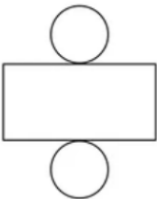
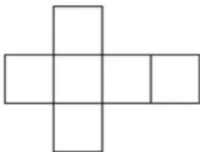
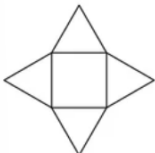
6. Encourage Sharing of Ideas. Select a few groups to share their examples and reasoning with the class. Acknowledge correct answers and clarify misunderstandings.
7. Bridge to Lesson Content. Summarize key takeaways from the activity. Use learners' examples to introduce the formal geometric terms (e.g., cube, cylinder, cone) and their surface features (flat, curved, faces, edges).

Part 3: 3D Object Hunt

Instructions

1. Explain the goal of the 3D Object Hunt: to identify real-life objects that resemble 3D geometric shapes.
2. Relate this activity to the lesson objective, understanding and describing 3D shapes and their surface characteristics.
3. Read aloud or project the instructions. Walk through the example in the template (e.g., "A shoebox is a rectangular prism with 6 flat surfaces"). Show a real object or image as a model to help learners visualize how to classify shape, surface type, and number of surfaces.
4. Facilitate Exploration and Collaboration. Allow learners to move around the classroom (if feasible) or brainstorm within their groups. Remind them to observe carefully and discuss their observations with group members. Encourage creativity (e.g., objects from home, stores, streets) while keeping the focus on shape features.
5. Monitor and Guide Group Work. Move around the room to observe group interactions. Provide prompts or questions such as:
 - a. "Does this object have only flat surfaces or both flat and curved?"
 - b. "Can you count and describe the surfaces of that shape?"
6. Offer support to groups struggling with identifying shapes or estimating surfaces.

	<table><tr><td>Ice Cream Cone</td><td>Cone</td><td>Curved, flat</td><td>2</td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr></table> Processing Questions: 1. Which objects were easiest to identify? 2. Which 3D shapes seem common in real life?	Ice Cream Cone	Cone	Curved, flat	2																					7. Invite groups to share a few examples from their template. Use a chart or board to list group findings and compare answers. Emphasize accuracy of geometric terms and clarify any misconceptions 8. Summarize the key features of the 3D shapes identified. Highlight the geometric terms: faces, surfaces, flat vs. curved. Transition into the formal lesson on properties of 3D shapes.
Ice Cream Cone	Cone	Curved, flat	2																							
	A.2. Establishing the Purpose of the Lesson Activity A.2: What Shape Am I? – Blueprint Challenge <i>Instructions:</i> 1. Identify 3D geometric figures (prisms, pyramids, cylinders, cones, and spheres) by analyzing their 2D blueprints (nets) or top, front, and side views. 2. You will be given a set of blueprints or views of 3D objects (these may be unfolded nets or top-front-side views). 3. Your task is to analyze the shape(s) in each blueprint or view and determine which 3D object it represents prism, pyramid, cylinder, cone, etc. 4. Match or write the correct 3D shape name beside each blueprint. 5. On your worksheet, sketch the 3D shape and then give the name of the 3D shape. Matching Table (Learner’s Worksheet):	Activity A.2: What Shape Am I? – Blueprint Challenge In this activity, learners will practice recognizing 3D geometric figures by studying their 2D representations. They will be given different nets (unfolded blueprints) or multiple views (top, front, side) of objects and will analyze these to determine the correct 3D figure. Learners will then sketch the solid, name it, and match it with the correct geometric term (prism, pyramid, cylinder, cone, or sphere). This activity develops spatial reasoning and strengthens the connection between 2D drawings and 3D solids. To facilitate this activity, follow these steps. <i>Instructions:</i> 1. <i>Start with a Question and Purpose</i> • Ask: “What do 3D shapes look like when unfolded into nets or viewed from the top, front, or side?” • Explain: This activity will help you connect 2D views to 3D objects. 2. <i>Show an Example</i> • Present a sample net or view (e.g., 1 rectangle + 2 circles).																								

Blueprint #	Sketch of the 3D Shape	Name of the 3D Shape	- Ask: “<i>What 3D shape could this form?</i>” (Answer: Cylinder). - Explain your reasoning aloud. 3. <i>Distribute Worksheets</i> - Give learners the nets or multi-view drawings with matching tables. - Ensure directions are clear (posted or written). 4. <i>Guide the Work</i> - Walk around and observe. - Ask guiding questions like: “<i>What shapes do you see?</i>” “<i>How many faces will it have when folded?</i>” “<i>What does this base tell us about the 3D shape?</i>” 5. <i>Share and Discuss</i> - Invite groups to present one match and explain their reasoning. - Show the blueprint on screen/board for whole-class feedback. 6. <i>Wrap Up and Connect</i> - Summarize: 2D views and nets help us understand 3D forms. - Reinforce key terms: <i>net, face, edge, base, vertex, prism, pyramid</i>, etc. - Bridge: “<i>Next, we’ll apply these ideas to calculate surface areas of 3D figures and solve real-life problems.</i>”
1. 		Rectangular Prism	
2. 			
3 			
4. 			
5. 			
6. 			

B. Instituting New Knowledge	B.1. Presenting Examples Activity B.1: Surface Area Stations Objective: Compute the surface area of various 3D shapes using the correct formula and measurements. Materials: 3D shape Measuring Device (ruler, tape measure) <i>Instructions:</i> 1. You will visit 5 different stations, each with a different solid shape. 2. At each station, you'll see: 3. A picture or model of the shape 4. Its dimensions (lengths, radius, height, etc.) 5. A surface area problem 6. Use a measuring device to determine the dimensions of the 3D shapes. Use the correct formula to compute the surface area. 7. Write your solution and final answer on your station worksheet. 8. Move to the next station when your teacher signals. Scenario per Station: Station 1: "Gift Wrapping a Toy Box" Scenario: You are wrapping a toy box as a birthday present. The box is shaped like a rectangular prism. Task Measure the dimensions of the toy box. Calculate the total surface area of the toy box to know how much wrapping paper you need.	Activity B.1: Surface Area Stations To facilitate the part of the lesson discussion, teacher may select any of the following strategies: <i>Before the Activity:</i> 1. Set up 5 stations with 3D shapes and measuring devices for: Rectangular Prism Cylinder Cone Sphere Square-based Pyramid 2. Prepare answer sheets and calculators. 3. Instruct the learners that they must determine the measures of the dimensions of the 3D shapes and record them on the station worksheet <i>During the Activity:</i> 1. Group learners (3–5 members per group). 2. For each 3D shape, allow the learners to do the measuring of the dimensions, record the measures obtained and calculate the surface area. 3. Rotate every 5–6 minutes or when the majority have completed the task. 4. Provide guidance or hints where learners struggle with formulas or units. 5. After the Activity: 6. Go over answers with the whole class. 7. Emphasize the logic behind each formula and common errors to avoid. Alternatively, instead of models, the stations can contain scenario cards that the learners will work on. In this case, just provide the scenario cards and calculators (no need to provide measuring devices.)
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Station 2: "Designing a Soda Can Label"

Scenario: You are helping design a new label for a soda can. The label must cover the entire surface area of the can (including the top and bottom).

Task:

Measure the radius and height of soda can.
Find the total surface area of the cylinder so you can determine the correct label size.

Station 3: Cone – "Ice Cream Cone Packaging"

Scenario: A company is creating a protective wrapper for an ice cream cone. You need to calculate the surface area of the cone (excluding the base, since it's open).

Task:

Measure the radius and slant height of the cone.
Calculate the lateral surface area of the cone.

Station 4: Sphere – "Painting a Ball"

Scenario: You are repainting a smooth decorative ball for a school project. You need to know how much paint will be required.

Radius: 6 cm

Task: Compute the surface area of the sphere to find how much surface needs to be painted.

Station 5: Square-Based Pyramid – "Designing a Tent"

Scenario: You are designing a tent shaped like a square-based pyramid. The surface includes the square base and four triangular faces.

Base side length: 5 m

Slant height: 8 m

Task: Calculate the total surface area of the pyramid to know how much fabric is needed.

	Station Worksheet <table><tr><th>Station #</th><th>3D Shape</th><th>Dimensions</th><th>Surface Area Formula</th><th>Computed Surface Area</th></tr><tr><td>1</td><td></td><td></td><td></td><td></td></tr><tr><td>2</td><td></td><td></td><td></td><td></td></tr><tr><td>3</td><td></td><td></td><td></td><td></td></tr><tr><td>4</td><td></td><td></td><td></td><td></td></tr><tr><td>5</td><td></td><td></td><td></td><td></td></tr></table> Formula Bank (Use each formula only once. Match it with the correct shape above.) A. $SA=6s^2$ B. $SA=2lw+2lh+2wh$ C. $SA=2\pi r^2+2\pi rh$ D. $SA=\pi r^2+\pi rl$ E. $SA=4\pi r^2$ F. $SA=b^2+2bl$ (where b = base, l = slant height)	Station #	3D Shape	Dimensions	Surface Area Formula	Computed Surface Area	1					2					3					4					5					
Station #	3D Shape	Dimensions	Surface Area Formula	Computed Surface Area																												
1																																
2																																
3																																
4																																
5																																
	B.2. Discussing the Concept Activity B.2: Surface Area Hunt Objective: Apply surface area knowledge to objects in your environment. Instructions: 1. Inside the classroom, name 3 real objects that resemble these shapes: prism, cylinder, cone, pyramid, or sphere. 2. For each object: 3. Identify the 3D shape it resembles. 4. Measure or estimate its dimensions. 5. Compute its surface area using the correct formula. 6. Sketch the object and label the measurements you used. 7. Submit your work with the sketches and surface area solutions.	Activity: Surface Area Hunt In this activity, learners will connect their knowledge of surface area to real objects found in their classroom environment. By identifying everyday items that resemble prisms, cylinders, cones, pyramids, or spheres, learners will practice measuring or estimating dimensions, applying the correct surface area formulas, and sketching the objects with labeled measurements. This task allows them to see how geometry applies to real-life situations while strengthening their computation skills. To facilitate this activity, follow these steps: Before the Activity: 1. Explain clearly what objects and data learners need to gather (object, shape type, dimensions, computation). 2. Provide a sample entry as a guide. After the Activity: 1. Review outputs for accuracy and real-life connections. 2. Highlight creative or accurate examples during class.																														

B.3. Developing Mastery

Activity: Surface Area Mastery Maze

Instructions:

1. You are trapped in a surface area maze!
2. To escape each “room,” solve the surface area problem in that room.
3. You must choose the correct formula, compute the surface area, and then use the answer to find the next correct room.
4. Each room will give you two possible doors (answers) - only one is correct!
5. Reach the final room to escape and earn your Geometry Master badge!

Maze Flow Sample (5 Problems)

Room #	3D Shape & Problem	Options to Exit	Correct answer
1	A cube with edge length 7 cm. What is its surface area?	A. 294 cm^2 B. 343 cm^2	A
2	A cylinder with radius 5 cm and height 10 cm. Surface area?	A. 471 cm^2 B. 235.5 cm^2	A
3	An open cone (no base) with radius 4 cm and slant height 9 cm.	A. 113.1 cm^2 B. 180.95 cm^2	A
4	A rectangular prism ($12 \text{ cm} \times 8 \text{ cm} \times 6 \text{ cm}$). What is the surface area?	A. 432 cm^2 B. 396 cm^2	A

Activity: Surface Area Mastery Maze

In this activity learners will firm-up their learnings by solving a sequence of surface area problems correctly to advance through a “maze” of rooms. Each room requires accurate computation and identification of the correct 3D shape and formula.

Before the Activity:

1. Print or prepare a maze handout or slide presentation.
2. Review formulas for each solid.
3. Create “checkpoints” or signs around the room for kinesthetic learners.

During the Activity:

1. Monitor learners’ progress and check accuracy.
2. Provide support as needed or create “hint cards” for struggling learners.
3. Use peer checking or allow learners to work in pairs.

After the Activity:

1. Review answers as a class.
2. Ask: “Which shape was most challenging to calculate?”
3. Clarify misconceptions about formula use or unit conversion.

	5	A sphere with radius 7 cm. What is the surface area?	A. 175.93 cm ² B. 615.75 cm ²	B	
C. Demonstrating Knowledge and Skills	C.1. Finding Practical Application Activity C.1: “Build It Right – Design a Mini House!” Objective: Learners will apply surface area formulas of 3D solids to calculate material needs (e.g., paint, wallpaper, roofing, etc.) in a practical design scenario. <i>Instructions:</i> - Scenario: You and your team are hired to design a mini dream house model. You need to calculate the total surface area of different parts of the house to know how much paint, wallpaper, tiles, and roofing materials are needed. - In groups of 3–4: - Choose from pre-designed 3D house structures (or create your own). - Break down the house into 3D components: rectangular prism (main walls), triangular prism (roof), cylinder (tower/chimney), etc. - Measure or use given dimensions of each part. - Compute the surface area for each component. - Total the surface areas to find how much material is needed (e.g., paint in m²). - Present your computations and budget to the class. - Materials Needed: Grid paper or cardboard cutouts Ruler, calculator, colored pens "Material Prices" Reference Sheet Sample Structure & Dimensions (Can be Modified)				Activity C.1: “Build It Right – Design a Mini House!” This activity is designed to be completed as group work. Ensure that all learners actively participate and collaborate effectively. This task is an opportunity for them to deepen their understanding and refine their skills, as it will help prepare them for the upcoming performance task. Encourage them to aim for thorough and high-quality work. Provide feedback on their output. In this part you will apply Experiential-Collaborative learning. <i>Instructions:</i> - Provide templates for 3D house parts or allow learners to build their own. - Guide learners in breaking complex shapes into simpler solids. - Monitor calculations and group work. - Facilitate final presentations or gallery walk.

Component	Shape	Dimensions (cm)	Material Used
Main House Body	Rectangular Prism	20 x 15 x 10	Paint
Roof	Triangular Prism	Base = 20, Height = 8, Length = 15	Roofing Tile
Chimney	Cylinder	Radius = 2 Height = 5	Brick surface

C.2. Making Generalizations

Activity C.4: Wrapping Up: Key Takeaways on Surface Area

Option 1: Math Reflections: What I Learned, What I Wonder

Instructions:

After completing the surface area lessons and problem-solving tasks, take 5–10 minutes to reflect individually. Use the following sentence starters to write your own learning statements and questions.

I realized that _____.

I found it challenging when _____.

I wonder why/ how _____.

Next time, I want to _____.

Be honest and thoughtful in your answers—you may be called to share your insights.

Option 2: Map the Math!” – Surface Area Concept Web

Instructions:

Activity C.4: Wrapping Up: Key Takeaways on Surface Area

For the summary, the approach to be used is Reflective Learning. To facilitate this part, you may select and facilitate one of the provided activities.

1. At the center of your paper, write the main concept:
"Surface Area of 3D Shapes"
2. From the center, draw branches for each major 3D shape:
Rectangular Prism
Cube
Cylinder
Cone
Sphere
Pyramid
3. For each shape, include:
A small sketch of the shape
The formula for surface area
Key parts (e.g., radius, height, base area)
An example of a real-life object
A note on surface type (e.g., flat, curved, both)
4. Add color, arrows, and connecting ideas (e.g., "both have circular bases" or "used in packaging design").
5. Share your concept web with a partner or small group.

Option 3: Exit Ticket on Surface Area

Instructions:

1. Multiple Choice – Concept Recall
Which of the following shapes has only curved surfaces?
A. Cube
B. Cylinder
C. Sphere
D. Rectangular Prism
2. Computation – Apply Formula
Find the surface area of a cube with edge length 4 cm.
Show your work and answer.
Formula used: _____
Solution: _____
Final Answer: _____ cm²
3. Reflection – Personal Connection
One thing I learned today about surface area is...
_____.

One question I still have is...

C.3. Evaluating Learning

Activity C.3: Checking What We Learned

Task 1: Identify and Match (5 pts)

Instructions:

Match each solid figure with the correct surface area formula.

Solid Figure	Formula Choices (Write the letter)
1. Cube	a. $2\pi r(r + h)$
2. Cylinder	b. $6s^2$
3. Rectangular Prism	c. $\pi r^2 + \pi rl$
4. Sphere	d. $2(lw + lh + wh)$
5. Cone	e. $4\pi r^2$

Task 2: Compute! (15 pts)

1. *Instructions:*

Solve the following. Show your solutions clearly.

- Find the surface area of a rectangular prism with length = 5 cm, width = 3 cm, and height = 4 cm.
- Find the surface area of a cylinder with radius = 7 cm and height = 10 cm. Use $\pi = 3.14$.
- A cone has radius 6 cm and slant height 10 cm. Find its surface area.

Task 3: Real-Life Word Problem (10 pts)

Instructions:

Solve and explain in a sentence.

You are wrapping a gift shaped like a cube with 8 cm edges. How much decorative paper do you need to fully cover the gift (excluding waste)?

Show your formula, computation, and final answer with units.

Task 4: Analyze & Explain (5 pts)

Activity C.3: Checking What We Learned

This activity must be completed individually by each learner. Its completion is essential for assessing their level of understanding of the lesson. Review their work promptly so you can provide timely and appropriate intervention for those who require additional support or remediation.

Instructions:

- Use as an end-of-lesson quiz or part of a summative test.
- Assess both computational accuracy and conceptual understanding.
- Review written explanations for depth of insight.

	Instructions Reflect and respond in 2–3 sentences. "Why is it important to know the surface area of objects in real life? Give one example and explain how surface area helps in that situation."	
	C.4. Additional Activities Activity C.4: Practice More, Learn More <i>Remediation</i> Improving Understanding about Surface Area of 3D Shapes (Solid) Activity 1: Surface Area Puzzle Match Objective: Match 3D solids to their surface components and formulas. Instructions: You'll receive a set of cut-out puzzle cards: 3D Shape name Surface Area formula Net (unfolded drawing) Real-life object example Your task is to match the 4 pieces per shape and tape them together. Check with a partner or your teacher for feedback. Activity 2: Build It & Label It Objective: Use paper nets to construct 3D shapes and label faces. Instructions: Cut and assemble paper nets (cube, rectangular prism, cone, cylinder). Count the number of flat and curved surfaces. On each face, write the shape name (e.g., rectangle, circle) and its area formula (e.g., $A = lw$). Use your labeled model to guide surface area computation. Activity 3: Formula Fill-in Practice Objective: Practice applying formulas step-by-step with scaffolding.	Activity C.4: Practice More, Learn More This additional exercise may be given as an assignment or homework. Remediation should be provided to learners who have not yet achieved mastery of the lesson, while enrichment activities should be offered to those who are ready for further learning. You may choose any one or two activities appropriate for the learners. <i>Instructions:</i> Provide extra guidance, visuals, and manipulatives. Use small-group instruction or peer pairing. Check for concept understanding through questions before reintroducing formulas. Note: Activity 1 may be given to learners to reinforce shape recognition, formula recall, and visual understanding of faces. Activity 2 is great for tactile and visual learners. Activity 3 builds fluency in formula use and arithmetic. For enhancement you may one or two from Tasks 1 to 5, whichever is appropriate for the learners. <i>Instructions:</i> Provide open-ended prompts and materials for hands-on work. Offer guidance but let learners take creative and computational risks. Encourage presentation or peer feedback for real-world connections. Pedagogies:

	Instructions: Use guided worksheets with partially filled formulas for surface area. Example: $SA = 2\pi r(r + h)$ Given: $r = \underline{\hspace{1cm}}$, $h = \underline{\hspace{1cm}}$ Step 1: Plug in the values Step 2: Multiply and simplify Final Answer: $\underline{\hspace{2cm}}$ cm^2 Repeat for different solids with increasing independence. Activity 4: “My Surface Area Story” – Real-Life Application Writing Objective: Connect surface area to everyday situations. Instructions: Choose one real-life object (e.g., box, can, ball). Describe the object’s 3D shape. Write how to compute its surface area and why it might be important (e.g., painting, wrapping). Draw and label the object if helpful. <i>Enhancement</i> Enhancement Tasks – Surface Area of Solid Figures Task 1: Design a Packaging Box Objective: Apply surface area knowledge in product design. Instructions: Imagine you are designing a box for a new product (snack, toy, gift, etc.). Decide the 3D shape of the box (e.g., cube, rectangular prism, cylinder). Determine the dimensions based on your product’s size. Calculate the surface area of the box. Design the outer layout (color, label, logo) that would cover the surface area. Task 2: Solve a Real-World Problem Combo	Project-Based Learning STEM Integration Constructivism + Real-Life Application Differentiation – Caters to diverse learning paces and interests Note: For task 1, an extension may be given such as comparing the cost of packaging different shape options. For task 2, the teacher may adjust the problem using different real-world combinations, like a cone on top of a cylinder, or a pyramid with a cube base. For task 3, the teacher may give bonus points for learners who swap problems with a classmate and solve each other’s problems. Task 4 can be done with modeling clay, paper nets, or recycled materials.
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	Objective: Compute total surface area of combined solid figures. Instructions: Analyze the following figure: a cylindrical water tank with a hemispherical top. Radius = 5 m, Height of cylinder = 12 m Find the total surface area (exclude base of hemisphere if attached to cylinder). Show all steps and formulas used. Task 3: Write a Surface Area Word Problem Objective: Create and solve your own word problem involving surface area. Instructions: Write a real-life scenario involving a 3D object (e.g., tent, swimming pool, lamp). Include dimensions and what the person needs to find (paint, wrap, cover). Solve your problem and present it to a partner or group for peer checking. Task 4: Math + Art Integration – Surface Area Sculpture Objective: Design a sculpture using 3D shapes and compute total surface area. Instructions: Create a model or sketch of a 3D "sculpture" using at least 3 different solid figures. Label the dimensions of each part. Calculate the total surface area. Decorate or explain how the surface area relates to covering the sculpture.	
Lesson 4.4		
III. CONTENT	Calculating the volume of prisms, pyramids, cylinders, cones, spheres and combinations of these. <i>(Recommended time: 2 hours)</i>	
IV. OBJECTIVES	By the end of the lesson, learners will be able to: 1. recognize and describe the components (e.g., base area, height, radius) needed to compute volume.	

	2. recall and use appropriate formulas for finding the volume of prisms, pyramids, cylinders, cones, and spheres. 3. calculate the volume of each solid figure using correct substitution and mathematical operations. 4. apply volume formulas to solve real-life and word problems, including composite figures.	
IV. PROCEDURES	ACTIVITIES	ANNOTATION
A. Activating Prior Knowledge	A.1. Activating Prior Knowledge Activity A.1: Volume Match-Up Relay <i>Instructions:</i> - Form groups of 4–5 members. - Each group receives a set of Volume Cards, including: - Cards with 3D shapes (name and sketch) - Cards with volume formulas - Cards with sample problems (with given dimensions) - Cards with the correct volume answers - Your task is to: - Match each 3D shape card with its corresponding volume formula. - Solve the sample problem using the formula. Find the correct volume answer card and match it with the problem. - Once done, your group checks the answers with the teacher. - The fastest group with correct matches wins a small reward!	Activity A.1: Volume Match-Up Relay To activate and assess learners’ prior knowledge on volume formulas of 3D solids through a collaborative and engaging matching activity. In facilitating this activity you may apply the following strategies. Instructions: <i>Before the activity:</i> - Prepare sets of Volume Cards for each group: 3D shape cards (name + sketch) - Formula cards - Problem cards (with given dimensions) - Answer cards (correct volume) - Organize materials by sets and label them clearly. - Prepare a simple prize for the winning group. - Introduction of the Activity: - Greet learners and state the objective: “Today, we’ll warm up by reviewing what you already know about finding the volume of different 3D solids.” - Briefly review the types of 3D shapes and volume formulas without solving problems yet. - Explain the mechanics of the “Volume Match-Up Relay.” - Group Formation and Material Distribution: - Divide the class into groups of 4–5 learners. - Distribute one complete set of cards to each group. - Ensure all groups understand the matching and problem-solving tasks. <i>Facilitate the Activity:</i> - Signal the start of the relay. - Circulate the room to monitor progress, answer questions, and check engagement.

		3. Encourage teamwork and time management. 4. Take notes on common misconceptions or errors for later discussion. 5. Checking and Debriefing: 6. Allow groups to bring completed matches to the teacher for checking. 7. Acknowledge the fastest group with all correct answers and give the reward. 8. Discuss common errors and clarify misunderstandings. 9. Reinforce correct volume formulas and strategies for identifying solid types.
	A.2. Establishing the Purpose of the Lesson Activity A.2: Why Volume Matters? Real-Life Connections <i>Instructions:</i> Watch & Reflect 1. Watch a short 2–3-minute video (or presentation) showing real-life scenarios involving volume, (e.g. filling a water tank, packing boxes, baking a cake, or designing containers). 2. Think-Pair-Share: <i>Think</i> a. Individually, write down one real-life situation where you think calculating the volume is important. b. Pair c. Share and discuss your ideas with a seatmate. <i>Share</i> 3. Volunteer or be chosen to share your group’s ideas with the class. 4. Answer the question: "Why do we need to learn how to calculate the volume of 3D solids?" 5. Write your response in 1–2 sentences on a sticky note or in your math journal.	Activity A.2: Why Volume Matters? Real-Life Connections This activity helps learners see why calculating volume is important in everyday life and prepares them to learn volume formulas for 3D shapes. <i>Instructions:</i> 1. Begin the lesson by saying: “Before we learn how to calculate volume, let’s see why it matters in real life.” 2. Present a short video or images of situations where volume is used (e.g., filling a pool, baking with measuring cups, packing boxes, designing water tanks). 3. Encourage learners to think by asking the following guide questions: ○ Where do you think volume is being used here? ○ Why do people need to measure volume in this situation? 4. Think-Pair-Share ○ Think: Learners recall a time they used or saw volume in real life. ○ Pair: Share their example with a partner. ○ Share: Selected pairs present to the class. 5. Conclude by asking:

		“Why do you think it’s important to learn how to calculate the volume of 3D solids?” Learners write a short personal response on a sticky note or in their math journal. 6. Collect and post responses (or summarize them on the board) to create a shared class understanding of the importance of volume.
B. Instituting New Knowledge	B.1. Presenting Examples Activity B.1: What’s Inside? Understanding Volume Conceptually Objectives: Develop conceptual understanding of volume as the amount of three-dimensional space occupied by an object, prior to introducing specific formulas. Instructions: 1. Observe different 3D objects (e.g., a box, bottle, cone-shaped cup). 2. Estimate which object can hold and explain your reasoning. 3. Join a guided discussion with your teacher on what <i>volume</i> means. 4. Compare it with <i>surface area</i> to understand the difference. 5. Develop your understanding by answer the following questions: ○ <i>What do we mean by “volume”?</i> ○ <i>How can we measure it?</i> ○ <i>What happens to the volume when we change the height or base of an object?</i>	Activity B.1: What’s Inside? Understanding Volume Conceptually This will help learners build an intuitive understanding of volume before applying formulas. To facilitate the discussion of the following topics you may apply any the following strategies: <i>Instructions:</i> 1. Present different 3D objects (e.g., shoebox, water bottle, ball, cone-shaped cup). 2. Ask: “Which of these do you think holds the most? The least? Why?” 3. Encourage learners to share guesses and reasoning based on observation. 4. Pose the central question: “What is volume, really?” 5. Let learners describe volume in their own words (e.g., “space inside,” “how much it holds”). 6. Collect answers on the board, then clarify with: “Volume is the amount of space inside a 3D object.” 7. Differentiate Surface Area vs. Volume using simple analogy: “Surface area is the wrapping paper around a box; volume is what fits inside the box.” 8. Reinforce with examples learners can relate to. Introduce cube units by asking: “How many little cubes could fit inside this box?” 9. Use fluid analogy: “How much water would fill this container?” 10. Ask guiding questions: ○ What happens to the volume if the box gets taller?

		○ Why do we multiply length, width, and height to get volume? 11. Allow learners to share examples from their experiences. 12. Invite questions and reflections to check understanding.
	B.2. Discussing the Concept Activity B.2: Volume Explorers: Discover the Formula Instructions: 1. Observe Carefully. Watch as the teacher demonstrates how to calculate the volume of a 3D solid (e.g., rectangular prism) using a guided example. 2. Take Notes. Record the formula, steps, and sample solution provided. 3. Apply the Pattern. With guidance, identify what changes in the formula and solution when a different solid is used (e.g., from prism to cylinder, then to pyramid, etc.). 4. Guided Practice. Solve one example problem per shape together with the teacher using step-by-step prompting. 5. Check Understanding. Respond to teacher's questions throughout the demonstration (e.g., "What does the 'h' in this formula stand for?" or "What's the first thing we need to do before we multiply?").	Activity B.2: Volume Explorers: Discover the Formula <i>Instructions:</i> 1. To introduce new knowledge, begin by explaining that today's focus will be on learning the specific formulas used to find the volume of different 3D solids. 2. Present a physical model or diagram of a rectangular prism and pose a thought-provoking question: 3. "If we wanted to fill this box with cubes, how would we know how many will fit inside?" 4. This concrete question will help learners visualize the concept of volume as space inside a solid. 5. Next, model how to calculate the volume of the rectangular prism using the formula $V = l \times w \times h$. 6. Solve an example problem on the board, clearly labeling all given dimensions, identifying each variable, and showing each step in the solution. 7. Work through the problem, then pause and ask learners guiding questions, such as: 8. "What unit will our answer be in?" or "Why do we multiply all three dimensions?" 9. This step-by-step modeling will make the procedure transparent and will encourage learners to think critically about the process. 10. After completing the first example, introduce a new shape, a cylinder and connect it to the previous example by comparing the structure of the formulas. Tell the learners, "Notice how the prism's formula multiplies the area of the base $\times$ height. The cylinder's formula does the same, but the base is a

		circle.” Then, introduce $V = \pi r^2 h$. Draw a cylinder, label the parts, and solve a sample problem. 11. This process is repeated for the pyramid, cone, and sphere, each time building on previously introduced concepts and emphasizing structural similarities and differences in formulas. 12. Throughout the demonstration, prompt learners to complete steps with them, asking questions like, “What’s the radius in this example?” or “Which formula fits this shape?” This interactive modeling ensures engagement and scaffolds learners toward independent problem-solving. 13. After each example, encourage learners to record the formula and key takeaways in a graphic organizer or formula table for easy reference. This organized approach aids retention and gives learners a tool they can return to during practice.
	B.3. Developing Mastery Activity B.3: Volume Drill and Skill Stations Learning Objective: Reinforce and deepen learners’ understanding and procedural fluency in using volume formulas for different 3D solids (prism, cylinder, cone, pyramid, sphere) through scaffolded and varied practice. <i>Instructions:</i> Rotate Through 5 Skill Stations - Each station focuses on one 3D solid. At each, you will: - Read a problem involving the shape. - Use the correct formula to compute volume. - Show complete solutions in your notebook. - Check your answer with the teacher or answer key after completing the task. - Get Feedback	Activity B.3: Volume Drill and Skill Stations To facilitate this activity the teacher may apply any of the following strategies. <i>Instructions:</i> - To develop procedural mastery, set up five learning stations, each dedicated to one type of 3D solid: rectangular prism, cylinder, cone, pyramid, and sphere. Before the rotation begins, briefly review the formulas and give clear instructions on how each station works. Learners are divided into small groups and begin rotating through stations in 3-to-4-minute intervals. - At each station, ensure that the provided problems increase in complexity starting with direct substitution and progressing to real-life contexts or slightly missing dimensions that require learners to interpret the problem. Circulate between stations, monitoring engagement, offering hints, clarifying misconceptions, and providing immediate, targeted feedback. This role

	vii. After each station, record your accuracy and any corrections on your Mastery Tracker sheet. viii. Finish the Challenge Task (Bonus) ix. At the final station, solve a mixed-shape word problem that involves finding the total volume of a composite solid. Station 1: Rectangular & Triangular Prisms Focus: Apply and interpret volume formula for prisms Formula: Rectangular Prism: $V = l \times w \times h$ Triangular Prism: $V = \frac{1}{2} \times b \times h \times l$ Tasks: Compute the volume: A box has a length of 10 cm, width of 6 cm, and height of 8 cm. ► What is its volume? Challenge Problem (Application): A water tank in the shape of a triangular prism has a base of 4 m, height of triangle 3 m, and length 5 m. ► How many cubic meters of water can it hold when full? Station 2: Cylinders Focus: Master volume of circular cylinders and applications Formula: $V = \pi r^2 h$ Tasks: Direct Problem: A cylindrical can has a radius of 5 cm and a height of 12 cm. ► What is its volume? (Use $\pi = 3.14$) Visual Reasoning: You double the radius but keep the height the same. ► How does the volume change? Explain briefly. Station 3: Cones	is essential in supporting learners while encouraging increasing independence. 3. To promote metacognition, provide each learner with a Mastery Tracker where they log their accuracy per station, note down which formula they used, and reflect on any corrections. Guide them to check their answers using either prepared answer keys or by discussing solutions together during feedback intervals. 4. For learners who complete their tasks early or demonstrate strong understanding, offer a “Challenge Task” at the last station: a composite solid problem combining at least two shapes (e.g., cone on top of a cylinder), promoting synthesis of learned concepts. Ensure differentiated support by pairing struggling learners with peer coaches. When appropriate, provide scaffolding hints using leading questions rather than direct answers. 5. After all stations are completed, conduct a short debriefing discussion, asking: ◦ Which formulas did you find easiest? ◦ Which problems were tricky and why? This reflection will help consolidate learning and prepare learners for independent practice or assessment. <i>Answer Key</i> Station 1: Prisms Rectangular Prism: $V = 10 \times 6 \times 8 = 480 \text{ cm}^3$ Triangular Prism: $V = \frac{1}{2} \times 4 \times 3 \times 5$ $= \frac{1}{2} \times 12 \times 5 = 6 \times 5 = 30 \text{ m}^3$ ◊ Station 2: Cylinders Volume: $V = 3.14 \times 5^2 \times 12$
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	Focus: Use and compare with the cylinder formula. Formula: $V = \frac{1}{3}\pi r^2 h$ Tasks: Direct Problem: A cone has a radius of 6 cm and a height of 9 cm. ► Find the volume of the cone. Compare with a Cylinder: A cone and a cylinder have the same base radius and height. ► What is the relationship between their volumes? Station 4: Pyramids Focus: Volume of square or rectangular pyramids Formula: $V = \frac{1}{3} \times \text{Base Area} \times h$ Tasks: Compute the volume: A square pyramid has a base of 10 m $\times$ 10 m and a height of 15 m. ► What is its volume? Word Problem: A sandbox is shaped like a rectangular pyramid with base 6 m by 4 m and height 3 m. ► How much sand (in m³) will fill it to the top? How do you define or measure the height of a pyramid? Station 5: Spheres Focus: Conceptual and computational fluency Formula: $V = \frac{4}{3} \pi r^3$ Tasks: Calculate the volume: A basketball has a radius of 12 cm. ► What is its volume? (Use $\pi = 3.14$) Application Challenge: A spherical balloon expands from radius 2 cm to 4 cm. ► By what factor does the volume increase?	$= 3.14 \times 25 \times 12 = 3.14 \times 300 = 942 \text{ cm}^3$ Reasoning (Radius Doubled): Volume increases by a factor of 4 because volume is proportional to r^2. ► Sample correct response: “If the radius is doubled, r^2 becomes 4 times greater, so the volume is 4 times as much.” ◊ Station 3: Cones Volume: $V = \frac{1}{3}\pi r^2 h = \frac{1}{3} \times 3.14 \times 6^2 \times 9$ $= \frac{1}{3} \times 3.14 \times 36 \times 9$ $= \frac{1}{3} \times 1017.36 = 339.12 \text{ cm}^3$ Relationship to Cylinder: The cone has one-third the volume of a cylinder with the same base and height. ► Sample correct answer: “The cone’s volume is 1/3 that of the cylinder.” ◊ Station 4: Pyramids Square Pyramid: Base area = $10 \times 10 = 100 \text{ m}^2$ $V = \frac{1}{3} \times 100 \times 15 = \frac{1}{3} \times 1500 = 500 \text{ m}^3$ Rectangular Pyramid (Sandbox): Base = $6 \times 4 = 24 \text{ m}^2$ $V = \frac{1}{3} \times 24 \times 3 = \frac{1}{3} \times 72 = 24 \text{ m}^3$ The height of a pyramid is the perpendicular distance from the apex (the topmost vertex) to the base. This is different from the lateral height which is measured along the triangular side of the pyramid and is used for the surface area. ◊ Station 5: Spheres Volume: $V = \frac{4}{3}\pi r^3 = \frac{4}{3} \times 3.14 \times 12^3$
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Bonus Challenge Station: Mixed Solids
Focus: Real-life applications and problem-solving.

Task:

A candle is shaped like a cylinder (radius 3 cm, height 10 cm) with a conical top (same radius, height 5 cm).

► What is the total volume of the candle?

Scoring Rubric (Per Station)

Criteria	4 Points	3 Points	2 Points	1 Point
Accuracy of Answer	Correct final answer with correct units	With minor mistakes in calculation or rounding	Significant error but correct setup	Incorrect or no attempt
Use of Formula	Correct and appropriate formula used	With minor error in the formula	Attempted but wrong or incomplete formula	No formula or incorrect method used
Labeling / Units	Proper use of units throughout	Minor unit omission or confusion	Units missing or inconsistent	No use of units
Explanation (if applicable)	Clear and logical explanation or comparison	Mostly clear with minor gaps	Vague or partially incorrect explanation	No explanation or completely incorrect

$$= \frac{4}{3} \times 3.14 \times 1728 = \frac{4}{3} \times 5427.12 \text{ cm}^3$$

$$= 7237.12 \text{ cm}^3$$

Volume Change (Radius Doubled):

Volume increases by a factor of 8 (since $V \propto r^3$).

► Sample correct explanation: "Doubling the radius makes the volume $2^3 = 8$ times greater."

◊ Bonus Challenge Station: Mixed Solids

Cylinder:

$$V = \pi r^2 h = 3.14 \times 3^2 \times 10$$

$$= 3.14 \times 9 \times 10 = 3.14 \times 90 = 282.6 \text{ cm}^3$$

Cone:

$$V = \frac{1}{3} \pi r^2 h = \frac{1}{3} \times 3.14 \times 3^2 \times 5$$

$$= \frac{1}{3} \times 141.3 = 47.1 \text{ cm}^3$$

Total Volume:

$$V = 282.6 + 47.1 = 329.7 \text{ cm}^3$$

C. Demonstrating Knowledge and Skills	C.1. Finding Practical Application Activity C.1: Design a Mini Storage Box Objective: Apply knowledge of volume to solve real-world problems by designing a container that meets specified volume requirements using appropriate formulas for 3D solids. <i>Instructions:</i> 1. Your goal is to design a container (box, cylinder, cone, or a combination) that can hold a specific volume of material, such as 1,200 cm³ of rice or 2 liters of liquid. 2. Select a 3D solid (or combination of solids) for your design. Justify your choice based on space efficiency, purpose, and ease of measurement. 3. Use the appropriate volume formula(s) to calculate dimensions that will give you the required total volume. Show all calculations clearly. 4. Draw your container to scale and label all parts with measurements.	Activity C.1: Design a Mini Storage Box <i>Instructions.</i> 1. This is a sample activity for the “Demonstrating Knowledge and Skills – Finding Practical Application” phase of the lesson. It focuses on enabling learners to apply their understanding of volume in real-world contexts. It is designed to be completed as group work. Ensure that all learners actively participate and collaborate effectively. This task is an opportunity for them to deepen their understanding and refine their skills, as it will help prepare them for the upcoming performance task. Encourage them to aim for thorough and high-quality work. Provide feedback on their output. 2. To facilitate this activity, begin by presenting a real-world scenario: 3. “Imagine you’re hired to design a rice container that can hold exactly 1,200 cubic centimeters. How would you do it?” 4. Use physical props (like rice, water bottles, or boxes) to make the scenario concrete and engaging. This step is crucial in helping learners see the value of mathematics beyond the classroom. 5. Next, explain the activity instructions, emphasizing that learners must apply volume formulas, choose a practical shape, and justify their design mathematically. Model an example design on the board—perhaps a cylinder—with given dimensions, calculating the volume step-by-step to show how dimensions influence capacity. This initial modeling supports learners who may need help getting started. 6. As learners work on their designs, circulate around the room, asking guiding questions like: “What shape are you choosing and why?” “How do you know your container will hold exactly that volume?”
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		“What might happen if your measurements are slightly off?” 7. Provide formative feedback, addressing misconceptions about units, formula use, or scale. If needed, facilitate small group conferences to troubleshoot challenges collaboratively. 8. Once learners have completed their sketches and explanations, you may set up a gallery walk where groups display their designs. Learners walk around, view peers’ work, and leave sticky-note comments or questions. This step allows learners to communicate mathematical reasoning and appreciate multiple ways of solving the same problem. 9. Conclude with a reflective discussion: “Where else might we apply volume in everyday life?” Responses are linked back to the lesson’s learning objectives to reinforce relevance and mastery.
	C.2. Making Generalizations Activity C.2: Wrapping Up: Key Takeaways on Surface Area Option 2: Volume Summary Circle <i>Instructions:</i> 1. Form small discussion groups of 4–5 members. 2. As a group, recall the formulas and problem-solving techniques you used for finding the volume of prisms, cylinders, cones, pyramids, and spheres. 3. Discuss similarities and differences among the formulas. For example: What parts of the formula stay the same? What shapes use fractions (like $\frac{1}{3}$)? How does the base area influence the formula? 4. Create a Generalization Poster: On a sheet of paper, summarize your group’s findings by:	Activity C.2: Wrapping Up: Key Takeaways on Surface Area Wrap up, guide learners through the provided summary activity.

	Writing general rules or patterns observed in the volume formulas. Providing at least one example per solid with its volume formula and a labeled sketch. Highlighting any shortcuts, tricks, or insights you discovered. 5. Class Sharing: Present your generalization poster to the class. Discuss as a class what common themes emerge from all groups. Option 2: Exit Ticket – Volume Reflection Prompt Type: Individual Written Reflection Focus: Metacognition and conceptual understanding Instructions for Learners: Answer the following in 3–5 sentences: What was the most important thing you learned about volume? Which formula or shape do you find easiest and why? What strategy helps you avoid errors when calculating volume?														
C.3. Evaluating Learning Activity C.3: Volume Mastery Quiz <i>Instructions:</i> Accomplish the given tasks. Part A: Identify the correct formula for each 3D shape (matching type). <table><tr><th>Column A</th><th>Column B</th></tr><tr><td>Cube</td><td>A. $V=\frac{4}{3}\pi r^3$</td></tr><tr><td>2. Rectangular Prism</td><td>B. $V = l \times w \times h$</td></tr><tr><td>3. Cylinder</td><td>C. $V = \frac{1}{3} \pi r^2 h$</td></tr><tr><td>4. Sphere</td><td>D. $V = \pi r^2 h$</td></tr><tr><td>5. Cone</td><td>E. $V = \frac{1}{2}bh \times l$</td></tr><tr><td>6. Square Pyramid</td><td>F. $V = s^3$</td></tr></table>	Column A	Column B	Cube	A. $V=\frac{4}{3}\pi r^3$	2. Rectangular Prism	B. $V = l \times w \times h$	3. Cylinder	C. $V = \frac{1}{3} \pi r^2 h$	4. Sphere	D. $V = \pi r^2 h$	5. Cone	E. $V = \frac{1}{2}bh \times l$	6. Square Pyramid	F. $V = s^3$	Activity C.3: Volume Mastery Quiz <i>Instructions:</i> 1. Prepare a written quiz that will evaluate learners' retention and accurate use of volume formulas. 2. Before administering the quiz, remind learners of the expectations and the importance of showing complete solutions. 3. After collecting the quizzes, use a rubric to assess accuracy, problem-solving process, and the clarity of explanations. Immediate feedback should be provided in the next session, and common errors are addressed through a brief group review. Teacher’s Sample Assessment Rubric
Column A	Column B														
Cube	A. $V=\frac{4}{3}\pi r^3$														
2. Rectangular Prism	B. $V = l \times w \times h$														
3. Cylinder	C. $V = \frac{1}{3} \pi r^2 h$														
4. Sphere	D. $V = \pi r^2 h$														
5. Cone	E. $V = \frac{1}{2}bh \times l$														
6. Square Pyramid	F. $V = s^3$														

7. Triangular Prism	G. $V = \frac{2}{3}\pi r^3$
8. Hemisphere	H. $V = \frac{1}{3}b^2h$
9. Composite Solid (Cylinder + Hemisphere)	I. $V = \pi(\frac{d}{2})^2h$
10. Cylinder (with diameter given)	J. $V = \pi r^2h + \frac{2}{3}\pi r^3$

Part B: Compute the volume of five solids (including prism, pyramid, cone, sphere, and a composite solid). Solve the following volume problems. Use $\pi \approx 3.14$ when necessary.

1. A rectangular prism has a length of 8 cm, width of 3 cm, and height of 5 cm. What is its volume?
2. A square pyramid has a base side of 6 m and a height of 10 m. Find its volume.
3. A cone has a radius of 4 cm and a height of 9 cm. Compute its volume.
4. A sphere has a radius of 7 cm. What is its volume?
5. A composite solid consists of a cylinder (radius = 3 cm, height = 10 cm) topped with a hemisphere of the same radius. Find the total volume.
6. You are packing chocolate cubes into a box-shaped container with dimensions 8 cm $\times$ 3 cm $\times$ 5 cm.
 - a) What is the volume of the container?
 - b) If each chocolate cube occupies exactly 1 cm³, how many cubes will fit perfectly in the box?
7. A museum is designing a pyramid-shaped display case with a square base measuring 6 meters per side and a height of 10 meters. Calculate the volume of the display case.
8. An ice cream company designs ice cream cones with a radius of 4 cm and a height of 9 cm.
 - a) How much ice cream (in cm³) will fill the inside of the cone?
 - b) If 1 cm³ of ice cream weighs 1.1 grams and the

Criteria	4 - Excellent	3 - Good	2 - Needs Support	1 - Minimal
Accurately estimates and calculates volume	✓	✓	Some errors	Many errors
Works cooperatively in group	✓	✓	Needs reminders	Rarely contributes
Makes connections between models and formulas	✓ Deep insights	✓ Basic links	Partial understanding	Needs guidance
Completes worksheet with explanations	✓ Thoughtful	✓ Complete	Incomplete	Minimal effort

cone weighs 5 grams, how much would a full cone weigh in grams?

9. Toothpaste Container (Sphere + Cylinder) A new toothpaste container is in the shape of a cylinder (height = 15 cm, radius = 4 cm) with a hemisphere at one end (radius = 4 cm). How much toothpaste can the container hold in total?
10. You are designing a decorative pillar with a cylindrical base (radius = 3 cm, height = 10 cm) and a hemispherical dome on top (same radius).
11. Calculate the total volume of the structure.
 - a. If the cement for the pillar costs P10 per cubic centimeter, how much will you spend for the pillar?
 - b. How would the total volume change if the dome were a cone with height 3 cm instead of a hemisphere?

Part C: Short explanation questions

Answer each question briefly but clearly.

A sphere, a cylinder, and a cone have the same radius. When placed side by side on a table, the three shapes have the same height. Compare the volumes of the three shapes to each other.

Disregarding the units of measure, when will the volume of a sphere be bigger than its surface area? Explain how you obtained your answer.

You are given some taho in a conical cup. You are given the option of using a different cup. The first cup will have double the radius but half the height of the original, while the second cup will have half the radius but double the height. Which cup will you choose and why?

Activity 2: Volume Problem Solving Challenge

Type: Group Performance Task

Focus: Real-life application, collaboration

Instructions for Learners:

Each group will:
Solve any 2 real-world problems involving composite solids.

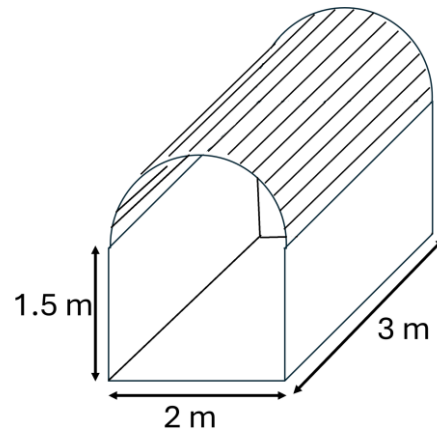
Problem 1: Water Tower Problem (Cylinder + Hemisphere)
A water tank consists of a cylinder with a hemisphere at the bottom. The height of the cylindrical part is 12 meters and its radius is 5 meters. What is the total volume of water the tank can hold when completely full? Use $\pi \approx 3.14$.

Problem 2: Candle Mold (Cone + Cylinder)
A candle mold consists of a cylindrical base with a radius of 3 cm and a height of 6 cm. On top of it is a conical section with the same radius and a height of 4 cm. What is the total volume of wax needed to completely fill the mold?

Problem 3: House Decoration (Rectangular Prism + Triangular Prism Roof)
A miniature model of a house is made of a rectangular prism (10 cm long, 6 cm wide, 8 cm tall) and a triangular prism roof (base = 6 cm, height of triangle = 4 cm, length = 10 cm). What is the total volume of the house model?

Problem 4: Juice Container (Sphere + Cylinder)
A novelty juice container is made of a cylindrical bottle (height = 15 cm, radius = 4 cm) with a hemispherical cap (radius = 4 cm). What is the total volume enclosed by the container?

Problem 5: Playground Design (Rectangular Prism + Half-Cylinder Tunnel)
A playground designer is building a tunnel. The base of the tunnel is a rectangular prism (length = 3 m, width = 2 m, height = 1.5 m), and on top of it is a half-cylinder (radius = 1 m, same length).



What is the total volume of the tunnel?

Present your solutions on a poster with:

Sketches of the shapes involved

Step-by-step computations

Final answers with units

Real-life explanation of the result

C.4. Additional Activities

Activity C.4: Practice More, Learn More

Instructions:

Read the problem carefully. The steps by step procedure may help you to better understand the lesson.

Step 1: Identify the 3D shape.

Hint: Look at the description—what is the solid shape mentioned?

My Answer: _____

Step 2: Write the correct volume formula.

Hint: Use the general volume formula for this shape.

My Answer: _____

Step 3: Draw and label the figure.

Activity C.4: Practice More, Learn More

This additional exercise may be given as an assignment or homework. Remediation should be provided to learners who have not yet achieved mastery of the lesson, while enrichment activities should be offered to those who are ready for further learning.

(Draw the figure and label the dimensions: length, width, and height.)

Step 4: Substitute values into the formula.

My Work:

Step 5: Solve.

My Answer: _____

Step 6: Answer the question with a complete sentence.

The volume of the water tank is _____.

Problems:

1. A water tank is shaped like a rectangular prism. Its length is 5 meters, its width is 2 meters, and its height is 3 meters. How much water can it hold?
2. A fish tank is 50 cm long, 30 cm wide, and 40 cm high. How much water (in cubic centimeters) can the tank hold when it is full?
3. A toy block is shaped like a cube with each side measuring 8 cm. What is the volume of the toy block?
4. A cylindrical water tank has a radius of 5 meters and a height of 10 meters. Find the volume of the tank.
A party hat is shaped like a cone with a radius of 6 cm and a height of 15 cm. How much space is inside the hat?

Enrichment:

Instructions

Read and solve each problem

1. A storage box is 2.5 m long, 1.8 m wide, and 1.2 m high. It is filled with smaller boxes, each measuring $0.3 \text{ m} \times 0.3 \text{ m} \times 0.3 \text{ m}$. How many small boxes can fit inside the storage box?

	2. A freezer produces ice cubes that are 5 cm on each side. If you want to fill a container with a volume of 2 liters, how many ice cubes are needed to fill the container? 3.. A conical water tank has a height of 12 m and a base diameter of 8 m. Water is being filled into the tank at a rate of 2 m³ per minute. Find the total volume of the tank. How long will it take to fill it completely? 4. A cylindrical silo has a radius of 4 m and a height of 15 m. If it is filled with grain and each cubic meter of grain weighs 720 kg: Find the volume of the silo. Find the total weight of the grain in kilograms. 5 water tank is made by attaching a hemisphere on top of a cylinder. The cylinder has a radius of 3 m and height 5 m. Find the total volume of the tank. •	
Lesson 4.5		
III. CONTENT	Converting Currencies and Time, and Using Scales in maps in Travelling Decision <i>(Recommended time: 2 hours)</i>	
IV. OBJECTIVES	At the end of the lesson, learners should be able to: 1. explain the concepts scales and maps, currency and time conversions, 2. interpret and use map scales to calculate real-world distances between locations on a map, 3. convert currencies and time zones across countries, 4. apply map scales, currency and time conversions across various contexts such as travel, real world distances, e-commerce, budgeting, and remittances, 5. apply map scales, currency, and time conversions when planning personal or group travels, 6. create a scaled map of a location or building to showcase its landmarks and attractions.	
V. PROCEDURES	ACTIVITIES	ANNOTATION
A. Activating Prior Knowledge	A.1. Activating Prior Knowledge	Activity A.1: Ratio World Trip

Activity A.1: Ratio World Trip

Instructions:

Solve the given problems, after solving, discuss answers as a class, highlighting any common errors or strategies.

1. On a road map, 1 inch stands for 10 kilometers. If the distance between two towns is 2.5 inches on the map, how many kilometers apart are they?

Solution: $1:10 = 2.5:x$, $x = 25$ km

The distance between two towns is 25 km apart.

2. A maja blanca recipe uses 2 cups of sugar for every 3 cups of flour. If you use 6 cups of sugar, how much flour do you need?

Solution: $2:3 = 6:x$, $x = 9$

3. Nine (9) cups of flour are needed.

After graduation, Earl John works abroad and sends \$600 to his parents each month. His parents divide the money in the ratio 4:3:3 for food, travel, and savings, respectively. How much of the remittance goes to each category?

Solution: $4x+3x+3x = \$600$, $10x=600$, $x=60$
 $4(60)+3(60)+3(60)=600$
 $240 + 180 + 180=600$

The remittance that goes to food is \$240, for travel is \$180, and for savings is \$180.

4. You are browsing in an online store which sells laptops and tablets in the ratio 3:5. If the store sold 200 devices in one month, how many of each were sold?

Solution: $3x+5x = 200$, $8x= 200$, $x=\frac{200}{8} = 25$
 $3(25) + 5(25) =200$, $75 + 125 =200$

The store sold 75 laptops and 125 tablets in one month.

This activity helps learners review ratio, proportion, and the metric system to reinforce mastery. To facilitate the activity, apply one of the provided strategies.

Materials Needed:

- Road map
- Calculator/smartphone
- Pen/pencil

Instructions:

1. Prepare five (5) problems written in separate papers.
2. Ask five learners to roll the papers in smaller pieces.
3. Divide the class into five groups.
4. Let the learners choose one representative in their group to pick one (1) problem that their group will discuss and solve.
5. Let each group discuss their problem and solve it by giving them appropriate time.
6. Ask one representative from each group to discuss their work highlighting any common errors and strategies.
7. Connect to the lesson by stating that their prior knowledge on ratio and proportion is important and pre-requisite for the lesson to be discussed on maps and scales.

Teaching notes:

- Many learners find application problems challenging.
- Encourage learners, whenever possible, to use algebra to solve problems even though they may be able to solve without it.
- The teacher may create another activity depending on the level of their learners.
- The teacher may also provide activity sheets for this topic.

	5. Grace receives a monthly allowance of ₱6,000.00. She decides to divide her allowance in the ratio 2:3:1 for savings, school expenses, and entertainment, respectively. How much money does Grace allocate to each category? Solution: $2x+3x+x=6000$, $6x=6000$, $x=\frac{6,000}{6} = 1000$ $2(1000) =2000$, $3(1000) =3000$, $1(1000) =1000$ Grace allocates ₱2000 for savings, ₱3000 for school expenses, and ₱1,000 PHP entertainment.	• This strategy is highly aligned with 21st-century skills like problem-solving, collaboration, and real-world application of academic content. Strategy 2; Problem Solving Relay Materials Needed: • Paper and pen/pencil • Calculator • Smartphone (conversion table application) • Set of conversion problems (written on index cards, paper or others) Instructions: 1. Divide the class into teams. 2. Each team gets a set of conversion problems. 3. Teams race to solve and pass the next question to the next teammate. 4. Use a check station where you can quickly validate answers before learners can move on to the next. 5. The group that finishes first wins. 6. Ask a representative from the winning group to explain their answers. 7. Connect the lesson by stating that, as we travel, we are going to need to convert one unit to another. Thus, the concept of metric units is of great help in our new topic. Teaching Notes: • Encourage learners to use a calculator, conversion table or conversion application online or offline using their smartphones. • Remind them that task distribution in a group work is highly encouraged to make work output fast. Every member of the group is important to make things work out. • Use a check station where you can quickly validate answers before learners can move on.
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	A.2. Establishing the Purpose of the Lesson Activity A.2: Wonders Without Borders Instructions: 1. Featured here are the photos of diverse tourist attractions from across the globe. 2. Identify the country to which each picture belongs to by stating the name of the country, and its official currency. Mount Fuji, Kyoto Temples, Tokyo Disneyland Answer: Japan, Yen – JPY - ¥ Great wall, Zhangjiajie, and Forbidden City Answer: China, Yuan – CNY ¥ Eiffel Tower, Palace of Versailles, Louvre Museum Answer: France, Euro – EUR € Venice canals, Colosseum, Leaning Tower of Pisa Answer: Italy, Euro – EUR € Statue of Liberty, Yellowstone, Grand Canyon Answer: United States, USD Dollar – US \$ Banff National Park, Niagara Falls, CN Tower, Beaches Answer: Canada, Canadian Dollar - CAD \$ Christ the Redeemer, Iguazu Falls, Amazon RainForest Answer: Brazil, Brazilian Real- R\$ Buenos Aires, Perito Moreno Glacier, Patagonia Answer: Argentina, Argentine Peso – ARG \$ Pyramids of Giza, Nile River, Luxor Answer: Egypt, Egyptian Pound - E£ Sahara Desert, Marrakech Medina, Atlas Mountains Answer: Morocco, Moroccan Dirham DH.	• This activity focuses on the fluency in metric unit conversion, Team collaboration and communication, confidence in explaining, mathematical reasoning, and engagement through fun, fast-paced learning. Activity A.2: Wonders Without Borders To facilitate this activity the teacher may apply one of the following strategies: Strategy 1: Immersive Simulation, Real-World Connection <i>Materials Needed:</i> • PowerPoint/Google Slides • Canva (for visually engaging slides • YouTube Playlist (featuring global tourist spots) • Google Earth (for interactive virtual travel) Instructions 1. Curate a presentation (slides, videos, virtual maps) showcasing famous tourist attractions across different countries. 2. Ensure each destination slide/video includes the name of a country, tourist spots, and national currency. 3. Start with enthusiasm and engagement: Say “Who wants to join me on a “World Tour” today?” 4. Create excitement by setting the tone for an imaginary travel experience. 5. Show the pictures, slides, or videos. 6. Briefly describe the attraction shown and highlight the country and its currency. 7. Ask “Can you guess which country this picture belongs to?” 8. Encourage student responses and discussion.
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	Sydney Opera House, Great Barrier Reef, Uluru Answer: Australia, Australian Dollar A\$ 3. Choose one of the countries listed above that you would like to visit soon. 4. Be prepared to share your reasons if asked. 5. Listen carefully as we connect this to today’s topic on travel.	9. Continue with multiple destinations to ensure engagement. 10. Personal Connection, ask learners: ○ Has anyone here visited another country before? ○ Who among you dreams of traveling abroad someday? 11. Say: “Some people travel, work, or live abroad for many reasons. Knowing how to convert time and currency between countries can be particularly useful, whether you are traveling, communicating with others, or making decisions about money.” 12. Introduce the main topic: Time and Currency Conversion 13. Connect their virtual experience to real-world application: “Let’s now learn how to convert time zones and currencies between countries - skills that travelers and professionals use everyday.” Strategy 2: Gamification (Flashcard Matching Game) <i>Materials Needed:</i> • Flashcards Instructions 1. Prepare the flashcards of the country’s name with its currency and famous tourist spots separately. 2. Ask the learners who among them want to visit other countries. 3. Set the Rules 4. Learners take turns flipping two cards per turn. 5. If the cards match, the student keeps the pair and another student takes another turn. 6. If not, both cards are flipped back, and the student’s turn ends, giving a chance to the next student. This will continue until all countries are matched with their tourist spots. 7. Ask learners: ○ Has anyone here visited another country before?
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		○ Who among you dreams of traveling abroad someday? 8. Say: “Some people travel, work, or live abroad for many reasons. Knowing how to convert time and currency between countries can be immensely useful, whether you are traveling, communicating with others, or making decisions about money.” 9. Introduce the main topic: Time and Currency Conversion 10. Connect their virtual experience to real-world application: “Let’s now learn how to convert time zones and currencies between countries - skills that travelers and professionals use everyday.” Teaching Notes: ○ Keep it visual and interactive – let the learners guess the locations and currencies ○ Encourage student participation by letting them share experiences or dream destinations. ○ Make it relevant – relate conversions to common real-life needs like travel, online shopping, or migrant work.
B. Instituting New Knowledge	B.1. Discussing New Concept Activity B.1: Unlock the Words: Building Strong Vocabulary <i>Key Terms and Definitions</i> Map - A visual representation of an area showing physical features, roads, cities, etc. Scale - The relationship between a distance on the map and the actual distance on the ground. Map Scale - A tool on the map that shows how distances on the map correspond to real-world distances. Bar Scale - (Graphic Scale) - A visual scale shown as a line or bar marked with distances.	Activity B.1: Unlock the Words: Building Strong Vocabulary To facilitate the discussion, employ the suggested steps. <i>Instructions:</i> 1. Prepare the materials needed: • Set of key terms written on different cards or pieces of paper • Matching definitions printed separately on corresponding cards or sheets • World Map 2. Distribute the set of key terms which are written on different cards or pieces of papers as well as their matching definitions.

Ratio Scale - (Representative Fraction)- A numerical scale written as a ratio (e.g., 1:100,000), meaning 1 unit on the map equals 100,000 of the same units in real life. Distance - The amount of space between two points, which can be measured on a map and converted to real-world measurement using the scale. Measurement - The process of finding the size, length, or amount of something, often using a ruler on a map in this context. Conversion - Changing a measurement from map scale to real-world distance using multiplication or division. Legend (Map Key) - A part of the map that explains the symbols and colors used. Currency- A system of money used in a particular country (e.g., US Dollar, Euro, Yen). Exchange Rate -The value of one currency in terms of another. <i>Example:</i> 1 USD = 0.90 EUR(may change anytime) Foreign Exchange (Forex)- The market where currencies are traded. The exchange rate is determined here. Base Currency - In a currency pair, it's the first currency listed. <i>Example:</i> In USD/EUR, USD is the base currency. Quote Currency (or Counter Currency)- The second currency in the pair. It shows how much of this currency is needed to buy one unit of the base currency. In USD/EUR, EUR is the quote currency. Time Zone - A region of the Earth that has the same standard time. Time zones are usually offset from	3. Mix them up and place them into separate piles. Teams race to match terms with their correct definitions. 4. Each group receives a passport sheet with missions involving key terms. The missions are listed below: Map Mission – Using a world map, measure the distance between Tokyo and New York using a ruler. Use the ratio scale to convert that into kilometers. Terms used: map, scale, ratio scale, distance, conversion Currency Checkpoint – You are traveling from the United States of America to France. If you have \$150.00 USD and the exchange rate is 1 USD = 0.90 EUR, how much will you have in Euros? Terms used: currency, exchange rate, base currency, quote currency Time Zone Task – You schedule a call at 3:00 PM Tokyo time. What time is it in New York (during DST)?” Terms used: time zone, UTC offset, time conversion, DST, local time Legend Challenge – Using a sample map, identify three features using the map key (legend) Terms used: legend (map key), measurement 5. Each team presents their solutions and explains which key terms they applied. 6. Encourage discussion: “What made this problem easy or challenging? How did using the scale or exchange rate help you?”
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Coordinated Universal Time (UTC) by a certain number of hours.

" **UTC (Coordinated Universal Time)**- The global time standard from which all other time zones are calculated. Example: UTC+0 is the reference point.

" **GMT (Greenwich Mean Time)**- An older term often used interchangeably with UTC, though UTC is more precise. GMT is based on the time at the Prime Meridian (0° longitude).

" **UTC Offset**- The difference in hours and minutes between a specific time zone and UTC.

Examples:

New York (Eastern Standard Time) = UTC-5

Tokyo (Japan Standard Time) = UTC+9

Daylight Saving Time (DST)- A seasonal adjustment where clocks are moved forward by 1 hour during warmer months to extend evening daylight.

Example:

Eastern Daylight Time (EDT) = UTC-4 (DST version of EST)

" **Local Time**- The actual time observed in a specific place, including any DST adjustments.

" **Time Conversion**- The process of calculating the local time in one time zone based on the time in another. Example: If it's 3 PM UTC, it's 11 AM in New York (UTC-4 during DST).

" **World Clock**- A tool or feature that shows the current time in different cities or time zones.

" **Time Zone Abbreviations**- Short forms for specific time zones. Examples:

EST – Eastern Standard Time (UTC-5)
PST – Pacific Standard Time (UTC-8)
CET – Central European Time (UTC+1)
IST – Indian Standard Time (UTC+5:30)

B.2. Presenting Examples

Activity B.2.1: Scales and Maps

Example 1.

You are planning a road trip from Manila to Baguio City, two popular cities in Luzon, Philippines. The map you are using has a scale of 1:1,000,000. You measure the distance between Manila and Baguio on the map as 22 cm. What is the real-world distance from Manila to Baguio City?

1:1,000,000 means 1 cm on the map = 10 kilometers in real life

Calculation using the scale: $22 \text{ cm} \times \frac{10 \text{ km}}{\text{cm}} = 220 \text{ kilometers}$

Answer: The real distance from Manila to Baguio City is 220 kilometers.

Example 2

You are looking for a hotel near the Eiffel Tower in Paris. You have seen on the map that there is a hotel which is 3 cm away from the Eiffel Tower. The map uses a scale of 1:5,000. Is the hotel within walking distance from Eiffel Tower?

1:5,000 means 1 cm on the map = 50 meters in real life

Calculation using the scale: $3 \text{ cm} \times \frac{50 \text{ m}}{\text{cm}} = 150 \text{ meters}$

Answer: The hotel is within walking distance since it is only 150 meters away from Eiffel Tower.

Example 3

Activity B.2: Scales and Maps

To facilitate this activity the teacher may apply the following steps.

Materials Needed:

- Physical or digital map of the Philippines, France, and Southeast Asia with a clear scale (e.g., 1:1,000,000, 1:5000)
- Ruler or digital measurement tool
- Calculator/smartphone (optional)
- Activity worksheet (with questions or tables)

Instructions:

1. The teacher will present sample scenarios that involve the use of scales and maps from various locations across the countries.
 2. Inform the learners that they will use a map of the Philippines to measure distances between key locations using the map scale and apply this knowledge to plan a realistic inter-island travel itinerary.
 3. Provide printed map scale or may use Google map online.
 4. Present to the class the Scenario:
 - “You are planning a road trip from Manila to Baguio City”
 - “You are looking for a hotel near the Eiffel Tower in Paris”
 - “Carlos is on a Southeast Asian cruise”.
 5. Facilitate the discussions
- Example 1.

Carlos is on a Southeast Asian cruise. The ship will travel from Manila, Philippines to HO Chi Minh City, Vietnam. On the cruise map, the distance between the two cities measures 12.5 cm. The map has a scale of 1:2,000,000. What is the real-world distance between Manila and Ho Chi Minh City based on the map? Scale on the map is 1:2,000,000, meaning 1 cm = 20 km in real-life Calculation using the scale: $12.5\text{ cm} \times \frac{20\text{ km}}{1\text{ cm}} = 250\text{ kilometers}$ Answer: The real-world distance between Manila and Ho Chi Minh City is 250 kilometers. Activity B.2.2: Budgeting and Time Travel Across Borders Currency Conversion <i>Example 1</i> Nicole is traveling from the United States to Europe. She has 500 USD and wants to convert it into Euros (EUR).	i. Locate the cities on the map: Manila and Baguio using a physical or digital map of the Philippines. ii. Use the map scale to calculate the distance (in kilometers) between each city. Illustration If 1 cm = 50 km, and Manila to Baguio is 4.8 cm in the map, then the actual distance is 240 km. iii. Estimate travel time: iv. Provide average speeds or travel times for each mode of transport. Example 2. i. Locate the nearest hotel to the Eiffel Tower using a physical or digital map of Paris. ii. Use the map scale to calculate the distance (in kilometers) from Eiffel Tower to the nearest hotel. iii. Estimate travel time using the walking distance. iv. Provide average speeds or travel times for each mode of transport. Example 3. Note: In this activity the skills and competencies that will be developed are cognitive skills (map reading and spatial logic, measurement, and critical thinking), real-world skills in decision making, problem-solving and planning as well as digital literacy (if Google maps used). The values that will be developed are appreciation of Philippine geography and other countries, collaboration and resourcefulness. Activity B.2.2: Budgeting and Time Travel Across Borders Here are the different strategies to facilitate the activity, you may apply any of these strategies.
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	The current exchange rate is 1 USD = 0.88 EUR. How much money in Euros will Nicole have? Given: Amount in USD: 500 Exchange rate (USD to EUR) on May 7, 2025: 1 USD = 0.88 EUR <i>Solutions:</i> Use the exchange rate to convert USD to EUR. Euros = USD x Exchange Rate Euros = 500 x 0.88 Euros = 440 Answer: Nicole will have 400 Euros <i>Example 2</i> Mr. De La Cruz prepares his trip from the Philippines to the United States. He needs to exchange Philippine pesos (PHP) for US dollars (USD). With PhP100,000, he wants to find out how much US dollars he will get based on the current exchange rate. Given: Amount in PHP: 100,000 Exchange rate (USD to PHP) on May 7, 2025: 1 USD = 55.42 PHP <i>Solutions:</i> To convert PHP to USD: Use the exchange rate to convert PHP to USD USD = PHP / Exchange Rate USD = 100,000.00 / 55.42 USD = 1,804.40 Answer: Mr. De La Cruz will get USD 1,804.40. This simple conversion can help him budget his expenses in the United States by knowing how much PHP is worth in USD before travel. <i>Example 3</i>	<i>Instructions:</i> 1. Begin the activity by saying: “You have seen different countries and their unique tourist attractions around the world. Now, let us explore how travelers plan their budget and manage their time when visiting these places. Through this, you will learn how to convert different currencies and adjust time zones between countries.” 2. Present situational examples related to travel budgeting and time planning, followed by thought-provoking questions to prompt engagement. 3. Guide learners in converting one currency to another using real or sample exchange rates. 4. Ask learners to convert the time from one country to another, reinforcing their understanding of time zone differences. 5. After presenting the examples, engage the learners with discussion questions such as: • How do we convert currencies? • What are the steps in converting time between countries? • Invite learners to think of a country they’d like to visit. Present this visually (using a PowerPoint slide or printed flashcards). 6. Ask guiding questions like: • Which country would you like to visit? • What is the exchange rate between Philippine pesos and that country’s currency? • Work through one sample problem together as a class, then let learners solve a similar example with a partner to reinforce the skill collaboratively. Strategy 2: Think-Pair-Share Technique <i>Instructions:</i> 1. Present one example problem related to time and currency conversion.
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Katrina wants to buy a dress from an online store in the US, where the price is 150 USD. How can Katrina convert the dress price to Philippine pesos? <i>Solutions:</i> 1. Search for the current exchange rate of USD to PHP (online or any other platforms.) (Supposing the current exchange rate of USD to PHP is 1USD =55.57 PHP) 2. Use the exchange rate to convert USD to PHP. 3. Get the given data. Given: 150 USD – price of the dress USD =55.57 PHP 4. Solve by converting 150 USD to PHP. Solution: 150 USD x 55.57 PHP Answer: The price of the dress in Philippine pesos is 8,335.50 PHP. <i>Time Conversion</i> <i>Example 1</i> Ariel lives in Albay, Philippines and he has a virtual job interview scheduled with a company based in New York City, USA. The interview is scheduled for 10:00 AM Eastern Time (ET) on December 10, 2025. What time and date should Ariel log in for the interview in Philippine Time (PHT)? <i>Solution:</i> 1. Time zone difference: Eastern Time (ET) is UTC-4 (during Daylight Saving Time), and Philippine Time (PHT) is UTC+8. So, the time difference is 12 hours with the Philippines ahead. 2. Convert ET to PHT: 10:00 AM ET + 12 hours = 10:00 PM PHT on Dec. 10. Answer: Ariel should log in at 10:00 PM Philippine Time on Dec. 10. <i>Example 2</i>	2. Ask learners to think silently about how to solve it, allowing them time to process independently. 3. Have learners pair up to compare and discuss their answers and reasoning. 4. Invite a few pairs to share their ideas with the class, encouraging them to explain their thought processes. To facilitate the discussion, teacher may select any of the following strategies: Strategy 1: Formative Game-Based Learning <i>Materials Needed:</i> • World map with a ratio scale • Currency Exchange table • Time zone chart or access to time zone converter • Worksheets with travel scenarios <i>Instructions:</i> 1. Inform the learners that they need to act as travelers solving real-world problems using maps, currency, exchange, and time zones. 2. Set up the activity by providing the materials needed for the activity. 3. Group the class into 5. 4. Let the learners know that there are five different scenarios written on small, folded pieces of paper.
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Lea is from Makati City, Philippines. She wants to call her mother who works in Dubai, United Arab Emirates (Gulf Standard Time – GST). She is free to call at 7:00 PM Philippine Time after doing her homework. What time will it be in Dubai when she makes the call?

Solutions:

1. Time difference:

Philippine Time (PHT) is UTC+8

Dubai Time (GST) is UTC+4

Therefore, Philippines is 4 hours ahead of Dubai

2. Convert PHT to GST

7:00 Pm PHT – 4 hours = 3:00 PM GST

Answer: When Lea calls at 7:00 PM in the Philippines, it will be 3:00 PM in Dubai.

Example 3

Criselda, an OFW nurse in Seoul, South Korea (Korean Standard Time – KST), wants to call her children in Bangkok, Thailand (Indochina Time – ICT). Criselda will call at 8:00 PM KST after her shift in the hospital where she works. What time will it be in Thailand by then?

Solutions:

1. Time zone difference:

South Korea (KST) is UTC+9

Thailand (ICT) is UTC+7

Therefore, South Korea is 2 hours ahead of Thailand

2. Convert KST to ICT:

8:00 PM KST – 2 hours = 6:00 PM ICT

Answer: The time in Thailand by then is 6:00 PM ICT

B.3. Developing Mastery

Activity B.3:

Applied Math Practice: Maps, Time, and Money

Each student (or group) will draw one at random, and the selected scenario will be the basis for their activity.

5. Ask a representative from each group to pick one paper at random.

6. Let them read the scenarios they picked.

Scenarios:

Map Scale & Distance

You are flying from Paris to Rome. Use the map scale to measure the distance between the two cities. How far is the actual distance in kilometers?

Currency Conversion

You exchange \$100 USD in Rome. If the exchange rate is 1 USD = 0.90 EUR, how much money in Euros do you get?

Time Zone Conversion

Your flight leaves Paris at 10:00 AM (CET) and lands in New York. What time is it in New York when you land if the flight is 8 hours long?

7. Using the map legend, identify a mountainous region on your route and explain what the symbol represents.

8. Create cards with time zone abbreviations and UTC offsets, and have learners match them to correct cities.

9. After the allotted time, have each group present their answers. They should explain which materials they used and describe the steps they followed to solve the given tasks.

Activity B.3:

Applied Math Practice: Maps, Time, and Money

	<i>Instruction</i> Solve the following: 1. You are planning a hiking trip from Point A to a campsite at Point B. On a topographic map with scale of 1:50,000, the measured distance between the two points is 7 cm. What is the real-world distance in kilometers? 2. You are given a road map of Mindanao with a scale of 1:1,200,000. The distance from Cagayan de Oro to Davao measures 18 cm on the map. What is the actual road distance from Cagayan de Oro to Davao? 3. You are in Manila (UTC+8) and need to schedule a meeting with a partner in Berlin, Germany (UTC+1). If it is 3:00 PM in Manila, what time is it in Berlin? 4. Your flight departs from New York (UTC-5) at 6:00 PM local time. What time do you arrive in Paris (UTC+1) 10 hours later? 5. You are travelling in Japan, and you want to buy a souvenir that costs ¥3,000. The exchange rate is 1 USD = ¥150. How much in US dollars will you spend? 6. You are given 500 Euro for a trip to France. If the current rate is 1 EUR =61.87 PHP 1.10 USD, how much is your budget in USD?	To facilitate this activity, you may select any of the following strategies: Strategy 1: Collaborative Learning, Critical Thinking Instructions: 1. Present the activity to the student: “City Distance Challenge” 2. Provide learners with a map of the Philippines with a scale of 1:500,000. 3. Let them choose 2-3 city pairs (e.g. Cebu to Tacloban, Davao to Cagayan de Oro). 4. Ask them to measure the map distance in cm. 5. Ask them to convert it using the scale. 6. Let them compare it with actual road distances using Google Maps and reflect on discrepancies (due to curves, road routes, etc.). Strategy 2: Active-Hands-on Learning Instructions: 1. Divide learners into small groups. 2. Assign each group a country (e.g. Japan, UK, India, Africa) 3. Provide a set of budgets in their local currency. 4. Give exchange rates and a list of typical travel expenses in the assigned country (e.g. hotel, food, transportation). 5. Steps: 6. Learners research exchange rates (or use provided ones) 7. Convert their budget into the assigned country’s currency. 8. Create a sample 3-day itinerary within budget. 9. Present and justify their budget choices to the class. 10. Pedagogy: Constructivist, Experiential Learning, Interdisciplinary Approach
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C. Demonstrating Knowledge and Skills	C.1.Finding Practical Application Activity C.1: Math Without Borders: A Travel Simulation The learners plan and simulate a multi-country journey involving time, distance, and currency calculations. <i>Instructions:</i> 1. Divide the class into groups (3-4 members) 2. Let them choose countries from different continents with different time zones and currencies. 3. Provided a world map with a scale, time zone chart, and currency exchange table. 4. Tasks: 5. Plot the travel route on the map and measure distances using scale. 6. Calculate flight durations and arrival times, adjusting for time zones and daylight-saving time where applicable. 7. Convert a given travel budget into each country's local currency using current exchange rates. 8. Create an itinerary including local activities, transportation times, and daily expenses. 9. Present the trip plan, justifying their time and currency decisions and highlighting the tools used. C.2. Making Generalization Activity C.2: Wrapping Up: Key Takeaways on Distance, Time, and Currency Option A. Travel Journal Reflection <i>Instructions:</i>	Activity C.1: Math Without Borders: A Travel Simulation This activity is designed to be completed through group collaboration. Ensure that all learners are actively engaged and contribute meaningfully to the task. It provides an opportunity for learners to deepen their understanding and strengthen their skills in preparation for the upcoming performance task. Emphasize the importance of producing thorough, high-quality work. As the facilitator, provide constructive feedback on each group's output. This activity incorporates an experiential-collaborative learning approach, where learners learn by doing and reflecting together. Activity C.2: Wrapping Up: Key Takeaways on Distance, Time, and Currency This activity encourages learners to reflect on what they learned and how they applied concepts related to map scales, time zones, and currency conversions to a simulated trip.

	1. Distribute a reflection journal prompt sheet (digital or paper). 2. Ask learners to write individually, then discuss in pairs or small groups. Reflection Prompts: <i>Map and Scale</i> • How did you use the map scale to calculate real distances? • What challenges did you face when converting map measurements into real-world values? • Time Zones • How did you determine the correct local time when planning your travel or communication between time zones? • Did daylight saving time or UTC offsets make it more difficult? How did you solve it? <i>Currency Conversion</i> • How did you convert your budget into different currencies? • What did you learn about fluctuating exchange rates and their impact on budgeting? • Application and Insight • Which of the three skills (scales, time, or currency) was the most challenging? Why? • How might these skills be useful in real life or future travels? Option B: Travel Team Debrief Circle <i>Instructions:</i> 1. In this activity you will reflect collaboratively on their problem-solving process, decision-making, geographic and mathematical skills as demonstrated in a real-world application group task or simulation.	For the summary phase, the most effective approach is Reflective Learning. To facilitate this, you may choose and guide learners through one of the suggested reflective activities provided.
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2. After completing a travel group activity – “Plan a Trip Around the World” and “Global Mission Passport”, you will sit in a circle with your group.

3. Each will receive a set of reflection question cards divided into three categories:

- Map and Scale
- Time Zone Calculations
- Currency Conversion Decisions

4. Each group selects a spokesperson to answer a card aloud. Then the group discusses their thought process and decisions behind it.

Reflection questions:

- How did your group measure distance, and did everyone agree on the results?
- Did you have any confusion on scales?
- What strategy helped you convert time most accurately?
- What was your budgeting approach when converting currencies?
- How would exchange rate changes affect your trip if this were real?

C.3. Evaluating Learning

Activity C.3: Checking What We Learned

Instruction:

Solve the following:

1. You are an environmental science student helping a local ecotourism company design a new hiking trail through a national park. You are given a topographic map with a scale of 1:50,000. The map shows that the proposed trail route, from the trailhead to the waterfall, measures 7.5 cm in a straight line. However, due to elevation changes and switchbacks, the actual trail will be 30% longer than

Activity C.3: Checking What We Learned

This activity is to be completed individually by each learner, as it plays a crucial role in assessing their understanding of the lesson. Review their outputs promptly to ensure timely and right intervention for learners who may need additional support or remediation.

the straight-line distance. Solve for the actual distance in kilometers.

2. You are a student preparing for a 10-day study program in Tokyo, Japan. Your budget is US\$1200, which you plan to convert into Japanese Yen (JPY). The current exchange rate is 1 USD = 145.50 JPY. You also need to consider a currency conversion fee of 2% charged by your bank. How much Japanese Yen will you have?
3. You are part of an international student team working on a virtual group project. The members are from: Manila, Philippines (UTC+8), Paris, France (UTC+2 during Daylight Saving Time), New York, USA (UTC-4 during Daylight Saving Time). You agree to hold an online presentation at 4:00 PM Paris time.
 - What time should the student in Manila log in for the meeting?
 - What time should the student in New York log in?
 - Create a scale model of the first floor of your senior high school building using a 1:100 cm scale.

C.4. Additional Activities

Activity C.4: Practice More Learn More

Remediation

Instruction:

Read the situations and answer the questions below.:

1. Anna is traveling from the Philippines to Japan. She wants to exchange her ₱5,000 Philippine Pesos into Japanese Yen (¥). The current exchange rate is 1 PHP = 2.10 JPY. How many Japanese Yen will Anna receive for her ₱5,000?

Activity C.4: Practice More Learn More

This additional exercise can be assigned as homework or an extension task. Provide targeted remediation for learners who have not yet demonstrated mastery of the lesson to help reinforce key concepts. For those who have already achieved proficiency, offer enrichment activities to extend their learning and challenge their higher order thinking skills.”

2. Maria is in Manila, Philippines (UTC+8). She wants to call her cousin in Dubai, United Arab Emirates (UTC+4). If it is 6:00 PM in Manila, what time is it in Dubai?
3. You are looking at a map of your province. The scale of the map is 1:200,000. You measure the distance from Town A to Town B as 7 cm on the map. What is the actual distance between Town A and Town B in kilometers?
4. You are using an emergency services map with a 1:250,000 scale. The fire station is 3.5 cm away from your location on the map. How many kilometers away is the station?
5. Luis is on a student exchange trip to the United States. He brought €100 Euros with him. At the currency exchange booth, the rate is: 1 Euro = 1.08 US Dollars. How many US Dollars (USD) will Luis get in exchange for his €100?

Enhancement

Instructions:

Read the situations and answer the questions below.

1. Alliah uses a world map with a scale of 1:20,000,000. She measures the distance between Manila and Berlin on the map and finds it is 31 cm. What is the real-world distance between Manila and Berlin?
2. Alliah has a travel budget of ₱50,000 Philippine Pesos. The current exchange rate is: 1 EUR = ₱60. How much will Alliah have in Euros (€) after converting her budget?
3. Her flight departs Manila at 10:00 PM(UTC+8) and takes 13 hours. She wants to inform her host in Berlin (UTC+1) of her arrival time. What time will it be in Berlin when she arrives?

Tasks:

	1. Calculate and explain how you used the map scale to find the distance. 2. Show the currency conversion from pesos to euros. 3. Determine the local arrival time in Berlin using time zone differences. 4. Reflect: What are the most important things to consider when traveling internationally?	
V. SUMMATIVE ASSESSMENT	For the summative, have learners complete the tasks on a separate page of this lesson exemplar.	
VI. REFLECTION	For the teachers, you may answer the following questions: 1. What went well in your teaching? 2. What challenges did you encounter? 3. How did you manage these challenges?	

UNIT TEST

Part I. Multiple Choice:

Directions. Choose the letter of the correct answer. In numbers 1 to 3, write the measurement in either scientific notation (for very large/small numbers) or standard form (for everyday-scale numbers), as appropriate.

- Which of the following items is approximately 1 centimeter in size?
 - The diameter of a dinner plate
 - The length of a baseball bat
 - The height of a smartphone screen
 - The width of a fingernail
- A newborn calf weighs about 40,000 grams. Which metric unit is most appropriate for expressing its mass?
 - kilograms
 - grams
 - milligrams
 - micrograms

◆ Express the mass 40,000 in scientific form: _____
- A swimming pool contains approximately 2.5×10^6 milliliters of water. Which metric unit is most appropriate to describe the total volume?
 - milliliters
 - liters
 - kiloliters
 - cubic meters

◆ Express the volume 2.5×10^6 in standard notation: _____
- The length of a marathon is about 42,195 meters. Which is the most appropriate metric unit?
 - centimeters
 - millimeters
 - kilometers
 - micrometers
- Two cones have the same slant height of 5 cm. Cone A has a radius of 3 cm, and Cone B has a radius of 4 cm. Which statement is true about their surface areas?
 - The surface area of Cone A is twice that of Cone B.
 - Cone A has a larger surface area because its height is greater.
 - Cone B has a larger surface area because its radius is greater.
 - The cones have the same surface area because the slant heights are equal.
- Which of the following best explains why the formula for lateral surface area of a cylinder is $2\pi rh$?
 - It gives the volume of the cylinder.
 - It represents the area of two circular bases.
 - It is derived from squaring the radius and multiplying by height.
 - It calculates the area of the curved surface unwrapped into a rectangle.
- What is the most appropriate reason to use scientific notation for very large or small numbers in science?
 - To confuse readers
 - To make the numbers longer
 - To avoid using decimal places
 - To simplify very large or very small values for easier understanding
- You're wrapping a spherical ornament with foil. The foil pack says it can cover about 600 cm^2 . The ornament has a radius of 7 cm. Based on your estimation; will one pack be enough?
 - Yes, with plenty left over
 - Yes, it will be just enough

- C. No, you'll need two packs
- D. It depends on the shape of the ornament.

9. A student in Sydney needs to join an online class scheduled at 10:00 AM Manila time. The student isn't sure about the time zone. What would be a reasonable way to know the correct time?

- A. Wait for the teacher to call
- B. Assume all countries have the same time
- C. Just join the meeting when it feels like 10 AM
- D. Check the time zone difference and adjust accordingly

10. A scale drawing of a school gym shows the length of a basketball court as 3 cm on paper. The scale is 1:50. Which is the best estimate of the real court's length?

- A. 1.5 meters B. 150 centimeters C. 15 meters D. 30 meters

Part II: Problem Solving

Directions: Consider the real-life scenario below and read each problem carefully. Then, choose the letter of the correct answer for each item.

Lakbay Math – A Journey Across the Philippines and Beyond

You are helping a balikbayan (returning Filipino from abroad) plan a well-deserved vacation and business trip across the Philippines and a few international destinations. The trip includes hiking, road travel in Mindanao, international meetings, flights, and even shopping for pasalubong (souvenirs). As the trip planner, you must use your math skills to:

- Read map scales
- Convert units
- Calculate time zones
- Manage money exchange rates

Task: Complete each scenario below and present your answers clearly in a travel journal or report format. Show your computations and explain your reasoning in Filipino or English.

1. Hiking in the Cordilleras

You are organizing a hiking trip for your guest in Sagada, Mountain Province. Using a topographic map with a scale of 1:50,000, you measure a trail from Lake Danum to Kiltapan Peak to be 7 cm long on the map. What is the actual distance of the hike in kilometers?

2. Road Trip in Mindanao

You're planning a road trip from Cagayan de Oro to Davao City using a road map with a scale of 1:1,200,000. The measured distance between the two cities is 18 cm. What is the actual distance in kilometers? This will help plan fuel and travel time.

3. Online Meeting Across Time Zones

Your balikbayan guest will attend a virtual meeting with a partner in Berlin, Germany (UTC+1). You are in Manila (UTC+8) and the meeting is scheduled for 3:00 PM Manila time. What time should the partner in Berlin be online?

4. Flight from the US to Europe

Your guest is flying from New York (UTC-5) to Paris, France (UTC+1). The flight departs at 6:00 PM New York time and lasts 10 hours. What time will the flight arrive in Paris local time?

5. Souvenir Shopping in Japan

Before returning to the Philippines, your guest stops in Japan and wants to buy a souvenir priced at ¥3,000. The current exchange rate is 1 USD = ¥150. Your guest only carries US dollars. How many US dollars will your guest need to buy the souvenir?

Answer Key:

Part I:

D. The width of a fingernail

A. kilograms, ♦ Scientific form: 4×10^4 grams

B. liters, ♦ Standard notation: 2,500 liters

C. kilometers

C. Cone B has a larger surface area because its radius is greater.

D. It calculates the area of the curved surface unwrapped into a rectangle.

D. To simplify very large or very small values for easier understanding

C. No, you'll need two packs

D. Check the time zone difference and adjust accordingly

C. 15 meters

Part II.

1. 3.5 kilometers

2. 216 kilometers

3. 8 AM Berlin time

4. 10:00 AM Paris time (next day)

5. 20 USD

PERFORMANCE TASK

SHS Student Learning Hub

As a senior high school student, you are part of a student council committee tasked to propose a "Learning Hub" — a multipurpose space where learners can study, collaborate, or relax during breaks. To present your proposal, you need to create a scale floor plan of your design using accurate measurements and appropriate scaling techniques.

Task:

1. Measure the dimensions of a real or proposed location in your school where the Student Hub could be placed (e.g., a spare classroom, open hall, or old library room).
2. Choose a scale (e.g., 1:100 or 1:50) and create a scale drawing of the floor plan on bond paper or graphing paper.
3. Include features such as tables, chairs, shelves, charging stations, or any furniture you'd expect in a student lounge — all proportionally scaled.
4. Label all dimensions clearly and provide a legend or key.
5. Write a short explanation (1 paragraph) on why your design is student-friendly and how using scale drawing helped in visualizing space

Reflect:

If the Learning Hub has a floor area of 10 meters × 8 meters and each study table measures 1.2 meters × 0.6 meters, how many tables can be placed in the room while still keeping the space comfortable and walkable for learners? Explain your answer.

Rubric:

Criteria	4 - Excellent	3 - Good	2 - Fair	1 - Needs Improvement
Scale & Accuracy 50%	All measurements are accurate and scaled.	Mostly accurate; minor errors in scale.	Several errors in measurements or scale.	Scale not followed or mostly incorrect.
Design/Layout 15%	Layout is neat, creative, and functional.	Design is clear and mostly well-planned.	Design is somewhat disorganized.	Design is unclear or incomplete.
Labels & Units 15%	All items are labeled with correct units.	Most labels and units are correct.	Some labels or units are missing/wrong.	Few or no labels and units shown.
Presentation 5%	Very neat, clean, and easy to understand.	Mostly neat and understandable.	Somewhat messy or unclear.	Hard to read or very messy.
Written Explanation 15%	Explains clearly how math helped in design.	Explanation is good.	Basic explanation, missing some points.	No or very weak explanation.

Score Interpretation:

18–20: Excellent Work

14–17: Good Understanding

10–13: Needs Some Improvement

Below 10: Try Again with Support